

Transport Barycenter-Guided Sample-to-Cluster Matching for Unaligned Multi-view Clustering

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Abstract—Multi-view clustering (MVC) relies on consistency learning to align and fuse multi-view information for building clustering decision boundaries. However, mainstream methods adopt sample-to-sample/distribution/structure similarity smoothing for consistency alignment, which builds upon continuous cluster manifolds with semantic and geometric overlap and reliable paired priors of sample correspondences. They suffer from semantic- and instance-level view-unaligned problems inherent in the real-world data with discrete non-convex structures and unreliable cross-view correspondences. Misled by unaligned noise, they weaken discriminative semantic boundaries via similarity smoothing over false-positive pairs, and degrade discrete non-convex structures by mistakenly bridging distinct cluster manifolds via pseudo-semantic interpolation. Such consistency alignment pursues highly similar representations, distributions and structures, which deviates from the clustering logic of many-to-one partitioning as well as its learning goal for separable semantic boundaries. Guided by the many-to-one principle, we jointly formulate consistency alignment and clustering decision as a novel sample-to-cluster map, termed **Multi-view Discrete Optimal Transport**. Specifically, **MvDOT** is instantiated as a cluster-level transport matching framework that first adopts semi-discrete OT to learn a global OT barycenter via aggregating semantic-geometric information from all views, and then transports samples to barycenter-anchored consensus clusters under consistency constraints on semantic assignment and geometric measure. Even with discrete cluster manifolds and unreliable sample correspondences, MvDOT achieves cross-view consistent alignment while preserving inter-cluster semantic boundaries to uncover the underlying cluster structures. Extensive experiments show **MvDOT** outperforms 10 baselines with higher confidence and stronger robustness in complex MVC tasks.

Index Terms—Multi-view Clustering, Semi-Discrete Optimal Transport Theory, Partially or Fully View-Unaligned.

1 INTRODUCTION

Multi-view clustering (MVC) benefits from multi-view information interaction and enhancement to capture cluster semantic boundaries. Under the data assumptions in Fig. 1(a) that continuous cluster manifolds with convex supports and reliable sample correspondences are shared across views, mainstream sample-to-sample methods adopt paired priors to construct instance- [1–5] or class-level [6–12] positive pairs and leverage continuous semantics to smooth paired samples into similar representations for cross-view consistency clustering [13]. However, real-world data generally exhibits two prominent characteristics: i) Discrete manifold distribution. Due to the numerical discreteness and semantic diversity of data features [14, 15], as well as the discrepancy in collection purposes and generation mechanisms across views, multi-view data presents typically non-convex and highly heterogeneous. Concretely, the non-convex data structure consists of multiple discrete sub-clusters separated by natural semantic boundaries. Affected by view heterogeneity, cross-view cluster manifolds with consistent semantics and structures are dispersed in the common space without semantic and geometric overlap [16–

18]; ii) Unreliable paired priors. Due to the complexity of data generation and collection, cross-view sample correspondences are absent or unreliable, which leads to partial or even full view-unaligned problems in MVC [19–21].

Multi-view data with “discrete cluster manifolds” and “unreliable sample correspondences” bring semantic- and instance-level view-unaligned challenges, forming the unaligned multi-view clustering (UMVC) task. Disturbed by unaligned noise, sample-to-sample representation smoothing is prone to erroneous alignment due to unreliable similarity measurement and pairwise construction. In Fig. 1(b), pseudo-semantic interpolation over false-positive pairs spuriously connects discrete manifolds distinct in semantics and structure, thereby blurring natural semantic boundaries and disrupting discrete non-convex structures. This issue stems from the objective mismatch between upstream alignment and downstream clustering tasks. Consistency alignment is supposed to provide discriminative semantic boundaries for the many-to-one discrete map in clustering partition [22]. In contrast, continuous smoothing focuses on representation similarity to reduce heterogeneity across views, yet neglects to preserve the intrinsic semantic gaps among discrete non-convex structures. Moreover, sample-to-sample methods is driven by pairwise relations and only enforces local consistency on limited cross-view correspondences [23–25]. Moreover, sample-to-sample methods is driven by pairwise relations and only enforces local consistency on limited cross-view correspondences. Therefore, recent studies on sample-to-distribution/structure alignment [26–34] discard paired priors and turn to modeling view-level semantic and structural consistency, which is more conducive to retaining discriminative semantics and separable structures. Regrettably, they rely on clus-

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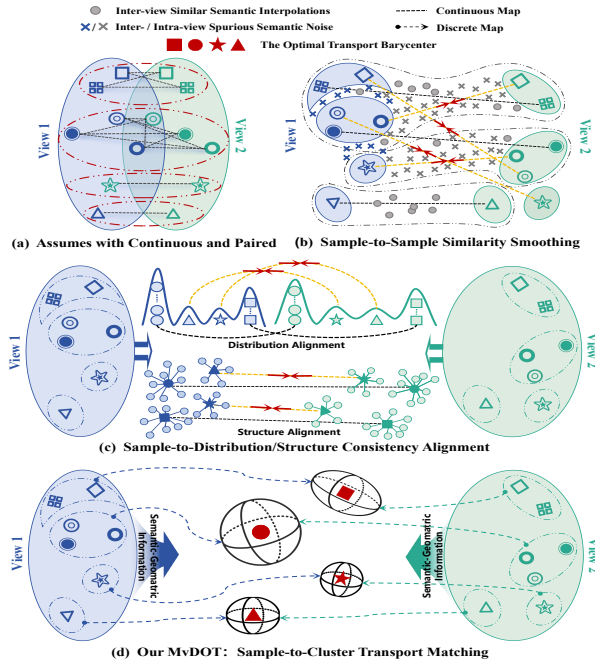


Fig. 1: Research Motivation. We present the method differences in the common space: Under ideal assumptions, (a) cluster manifolds are continuous with semantic and geometric overlap and cross-view observations correspond to the same instances. In real scenarios, view-heterogeneous manifolds is dispersed and cross-view correspondence is corrupted, (b) Sample-to-sample methods induce pseudo-semantic interpolations through continuous smoothing on false-positive pairs mismatched by unaligned noise, which forcibly connects semantically inconsistent manifolds into spurious manifolds; (c) Sample-to-distribution/structure methods, without the guidance of global consistency information and shared alignment anchors, induce biased estimation and constraints on cross-view distributional/structural alignment, and trap them in local consistency with semantic confusion and structure distortion; (d) Our sample-to-cluster map learns the barycenter via integrating semantic-geometric information of all data to anchor cluster regions in common space and unbiasedly guide samples toward their shared cluster, forming well-separated structures.

ter manifolds with continuous semantics and convex structures for reliable alignment, and remain confined to dual-view consistency due to lacking cross-view shared latent space or alignment anchors for parallel and unbiased information interactions [35–38]. Consequently, for UMVC tasks, existing methods fail to achieve unbiased global consistency across views and suffer from cluster degradation with ambiguous boundaries and loose structures.

To overcome these limitations, we extend the Semi-Discrete Optimal Transport (OT) theory to the multi-view setting and propose the Multi-view Discrete OT (MvDOT) method, which unifies consistency alignment and clustering decision into an innovative sample-to-cluster discrete mapping: without any view-aligned priors, it combines semantic similarity and topological connectivity as matching criteria, integrates all view semantic-structural information to learn a global OT barycenter with cluster manifolds shared across views, and transports all samples to their consensus clusters for cross-view consistent matching and clustering. Specifically, MvDOT is instantiated as a transport matching framework to handle the unaligned interference from discrete manifolds and unpaired samples: All view data are encoded into a common subspace, where multi-view OT maps are adopted to capture the semantic-geometric center of all data as the OT barycenter. For discrete manifolds, as the global

transport bridge of view semantic-geometric information, the OT barycenter induces shared cluster regions with mutually exclusive semantics and discrete non-convex structures, and guides discrete manifolds toward their consensus clusters with consistent semantic and structure. For unpaired samples, as an unbiased matching reference shared across views, the OT barycenter models global semantic and local topological constraints to discover cluster-level sample correspondences with minimal semantic-topological discrepancies. Benefiting from sample-to-cluster transport matching guided by the OT barycenter, MvDOT promotes full-view parallel interaction and enhancement to achieve global consistency alignment and unbiased clustering partition simultaneously, which mitigates semantic boundary blurring caused by similarity smoothing and prevents cluster structure degradation from unaligned noise interference.

The main contributions are summarized as follows:

- **Theoretical Innovation:** To our knowledge, this work is the first to systematically discuss and formally define view-unaligned dual challenges from discrete cluster manifolds and unreliable paired priors. Inspired by semi-discrete OT, we propose a sample-to-cluster transport matching and provide theoretical guarantees for its effectiveness and robustness in alignment and clustering.
- **Methodological Innovation:** MvDOT is instantiated as a cluster-level transport matching framework to handle discrete manifolds and unpaired samples, where an OT barycenter is adaptively learned as the shared anchor to guide alignment and fusion of discrete manifolds with similar semantic and isomorphic structure, while modeling global semantic and local topological consistency for exploring sample correspondences with intrinsic affinity.
- **Experimental Validation:** Extensive experiments on six representative datasets demonstrate that MvDOT comprehensively outperforms ten baselines under three unaligned settings. Notably, free from continuous distribution or paired prior assumptions, MvDOT breaks through baseline limitations of local consistency on partial view data and achieves globally consistent clustering across all views.

2 RELATED WORK

2.1 Unaligned Multi-view Clustering

Mainstream UMVC methods rely on sample-to-sample consistency alignment, in which limited pairwise priors guide the inference of cross-view relations [39, 40]. The seminal work PVC [41] employs a differentiable Hungarian to achieve instance-level optimal matching under pairwise supervision. Due to instance-level similarity suffering from false-negative noise, subsequent works [11, 42–44] design false-negative detection and correction mechanisms to explore semantic-level consistency. Nevertheless, driven by local consistency of positive pairs [45], sample-to-sample methods are sensitive to unreliable correspondences, which leads to erroneous cross-view alignment under unreliable similarity measurement [28] induced by semantic bias across views.

To alleviate the dependence on paired priors and pursue high-order consistency, sample-to-distribution/structure methods have been proposed: all views are assumed to share consistent distributions and isomorphic structures, and consistency constraints on distributions or structures are formulated for cross-view alignment. Methods such as SMILE [26] and CPMN [32] first model

view-specific distributions and prototypes, and then reduce cross-view distribution discrepancies or prototype distances via continuous smoothing. Without the guidance of reliable paired priors or shared alignment anchors, most UMVC methods perform biased alignment and clustering over dual-view semantic distributions or topological graphs, which renders them vulnerable to false-positive noise along with semantic confusion and structural degradation. [18] further attempts to learn the latent semantic distribution as a shared alignment anchor and explores a high-confidence view to guide reliable clustering. Overall these methods generally fail to unify consistency alignment and clustering decision under the many-to-one principle, which is more effective for preserving semantic boundaries and recovering cluster structures, and instead pay more attention to instance-level representation similarity or view-level distribution/structure consistency.

2.2 Optimal Transport for Multi-view Clustering

Optimal Transport (OT) finds optimal correspondences by minimizing transport costs between view distributions, which provides theoretical support for discrepancy measurement and semantic alignment across views [46]. With the development of entropy-regularized OT and efficient Sinkhorn solvers [47], OT has been widely incorporated into differentiable MVC frameworks. Unlike instance-level methods limited to local consistency on pairs, OT achieves implicit distribution-level alignment via the transport map without explicit correspondences or shared information assumptions. Thus, OT is increasingly adopted to mitigate semantic discrepancies and explores optimal correspondences across views [28, 31, 36, 48–50]. For instance, DHOT [51] conducts intra-view clustering to obtain view-specific cluster structures and minimizes the intra-cluster sliced Wasserstein distance between views to explore cross-view consistent semantics. Further study [52] finds that OT map can maintain geometric structures, which inspires OT-Hypergraph to model high-order structural relations and extend OT from distribution-driven semantic alignment to structure-aware neighborhood consistency. The early work GWMAC [48] exploits OT-based Gromov-Wasserstein [53] to aggregate all view structure information into a consensus kernel matrix for structural consistency alignment, which shows OT's potential in UMVC task.

Most OT-based methods follow sample-to-distribution mechanism, which merely reduces reliance on paired priors but still performs dual-view alignment in turns. They struggle to process all views in parallel for global consistency when more than two views are available [36, 51]. Moreover, classical OT is built upon continuous measures. Once applied to discrete non-convex datasets with diverse semantics, continuous transport interpolates noise into semantic gaps, which weakens intrinsic semantic boundaries and connects disjoint discrete regions [14, 54, 55]. To address this issue, semi-discrete OT [56] introduces a discrete measure via Dirac representation that decomposes continuous target distribution as $\nu = \sum_{k=1}^K \pi_k \delta_{c_k}$ with supports $\text{supp}(\nu) = \{c_k\}_{k=1}^K$. The transport map is restricted to the finite targets $T: \Omega \rightarrow \{c_k\}_{k=1}^K$. Based on Brenier's theorem [57], the discrete OT map is induced by a convex potential function $u(x) = \max_{k \in \{1, \dots, K\}} \{\langle c_k, x_k \rangle + h_k\}$ that is piecewise affine to partition the measure space into disjoint regions $\mathcal{W}(c_k) = \{x_k \mid \langle c_k, x_k \rangle + h_k \geq \langle c_l, x_k \rangle + h_l, \forall k \neq l\}$. Thus, semi-discrete OT models a many-to-one mapping from samples to centroids $T(x_k) = c_k, x_k \in \mathcal{W}(c_k)$. Such property integrates clustering partition into consistent learning, providing a theoretical foundation for exploring higher-level cluster alignment.

3 MvDOT: SAMPLE-TO-CLUSTER MAP

In this section, we first formally define the dual misalignment arising from discrete cluster manifolds and unreliable sample correspondences. Then, we propose MvDOT, a novel sample-to-cluster discrete map that extends semi-discrete OT to the multi-view setting, to prevent semantic structure degradation caused by unaligned noise. Finally, by combining OT theory with measure theory, we provide in-depth analysis and rigorous theoretical guarantees for the superiority of MvDOT.

3.1 Problem Statement

Notations. Given a multi-view dataset $\mathcal{X} = \{\mathbf{X}^v \in \mathbb{R}^{N \times d_v}\}_{v=1}^V$ with N instances across V views, $\mathbf{X}^v = \{\mathbf{x}_1^v, \mathbf{x}_2^v, \dots, \mathbf{x}_N^v\} \in \mathbb{R}^{N \times d_v}$ denotes the instance set of the v -th view, where d_v is the feature dimension. $\mathcal{Z} = \{\mathbf{Z}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$ and $\mathcal{H} = \{\mathbf{H}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$ denote the latent representations of $\mathcal{X}$ in the view-specific subspaces $\{\Omega^v \subset \mathbb{R}^D\}_{v=1}^V$ and the view-shared subspace $\Omega \subset \mathbb{R}^D$, respectively. The MVC task aims to partition N samples into K exclusive semantic clusters $\{\mathcal{C}_k\}_{k=1}^K$, where $\mathcal{C}_k \subseteq \{1, \dots, N\}$, $\mathcal{C}_k \cap \mathcal{C}_l = \emptyset, \forall k \neq l$. For each view v , the view-specific cluster manifold is defined as $\mathcal{S}_k^v = \{\mathbf{x}_i^v \mid i \in \mathcal{C}_k\}$.

Definition 1. Semantic Misalignment from Discrete Manifold Distributions: Hausdorff distance $d_H(\cdot, \cdot)$ quantifies the discrepancy between the supports of manifold distributions defined over non-empty compact sets in the common space:

$$d_H(\mathcal{A}, \mathcal{B}) = \max \left\{ \sup_{a \in \mathcal{A}} \inf_{b \in \mathcal{B}} \|a - b\|, \sup_{b \in \mathcal{B}} \inf_{a \in \mathcal{A}} \|a - b\| \right\}, \quad (1)$$

Given a dataset $\mathcal{X} = \{\mathbf{X}^v\}_{v=1}^V$ that satisfies

$$\mathcal{X}^v = \bigcup_{k=1}^K \mathcal{S}_k^v \text{ with } \text{int}(\mathcal{S}_k^v) \cap \text{int}(\mathcal{S}_l^v) = \emptyset, \forall k \neq l, \quad (2)$$

where $\mathcal{S}_k^v \subset \Omega$ is a non-empty compact set and $d_H(\mathcal{S}_k^m, \mathcal{S}_l^n) > 0$.

Semantic-level view misalignment arises from the discreteness of data distributions in the following two aspects:

(1) Intra-view non-convexity of semantic distribution: $\mathbf{X}^v$ is non-convex with diverse semantic structures,

$$\forall i \neq j, \exists \mathbf{x}_i^v, \mathbf{x}_j^v \in \mathcal{S}_k^v, \exists \lambda \in (0, 1), \text{ such that } \lambda \mathbf{x}_i^v + (1 - \lambda) \mathbf{x}_j^v \notin \mathcal{S}_k^v. \quad (3)$$

which implies the convex combination of intra-cluster different instances falls outside this cluster, reflecting intrinsic semantic-geometric disparities among distinct clusters.

(2) Inter-view discreteness of heterogeneous manifolds: cross-view manifolds $\{\mathcal{S}_k^v \mid k = 1, \dots, K\}_{v=1}^V$ where $\mathcal{S}_k^v = \{\mathbf{x}_i^v \mid i \in \mathcal{C}_k\}$, are discretely distributed in the common space,

- Intra-cluster manifolds exhibit geometric separations:

$$\forall k, \forall m \neq n, d_H(\mathcal{S}_k^m, \mathcal{S}_k^n) > 0. \quad (4)$$

View heterogeneity disrupts the formation of continuous cluster manifolds with semantic overlap in common space.

- Inter-cluster manifolds exhibit semantic exclusivity:

$$\forall k \neq l, \forall m \neq n, d_H(\mathcal{S}_k^m, \mathcal{S}_l^n) > 0. \quad (5)$$

cluster disjointness maintains intrinsic semantic boundaries without geometric overlap in common space.

Definition 2. Instance Misalignment from Unreliable paired priors: A dataset $\mathcal{X} = \{\mathbf{X}^v \in \mathbb{R}^{N \times d_v}\}_{v=1}^V$ is *view-unaligned* at the instance level if it contains two disjoint subsets: an aligned subset $\mathcal{D}_a = \{\mathbf{A}^v \in \mathbb{R}^{N_a \times d_v}\}_{v=1}^V$ with N_a paired instances and an unaligned subset $\mathcal{D}_u = \{\mathbf{U}^v \in \mathbb{R}^{N_u \times d_v}\}_{v=1}^V$ with N_u unpaired instances, where $\mathcal{D}_a \cap \mathcal{D}_u = \emptyset$ with $N = N_a + N_u$ and for any instance $\mathbf{x}_i^v \in \mathcal{D}_u$,

$$\sum_{m=1}^V \sum_{\substack{n=1 \\ n \neq m}}^V \mathbb{I}(\mathbf{x}_i^m, \mathbf{x}_i^n) = 0, \quad \forall i \in \{1, \dots, N_u\}, \quad (6)$$

where $\mathbb{I}(\mathbf{x}_i^m, \mathbf{x}_i^n) = 1$ if cross-view instances $\mathbf{x}_i^m$ and $\mathbf{x}_i^n$ correspond to the same sample $\mathbf{x}_i$, and 0 otherwise.

3.2 Multi-view Extension of Semi-Discrete OT

3.2.1 Multi-view Discrete OT Map

Given a multi-view dataset $\mathcal{X}$, we formulate MvDOT as follows:

- **Multi-view Source Distributions** $\mathcal{U} = \{\mu^v\}_{v=1}^V$: view data $\mathbf{X}^v$ induce absolutely continuous measures $\{\tilde{\mu}^v\}_{v=1}^V$ supported on Ω^v . Via Dirac decomposition, $\tilde{\mu}^v$ is discretized into the empirical measure $\mu^v = \sum_{i=1}^N a_i^v \delta_{\mathbf{x}_i^v}$, where $\delta_{\mathbf{x}_i^v}$ is the dirac measure at the i -th instance $\mathbf{x}_i^v$ and $\{\mathbf{a}^v = (a_1^v, \dots, a_N^v)^\top \mid a_i^v \geq 0, \sum_{i=1}^N a_i^v = 1\}_{v=1}^V$ is the source mass to be transported from each instance.
- **View-shared Target Distribution** ν : A discrete measure $\nu = \sum_{k=1}^K \pi_k \delta_{\mathbf{c}_k}$ supported on Ω , serves as the target measure for source distributions $\{\mu^v\}_{v=1}^V$. Dirac atoms $\{\delta_{\mathbf{c}_k}\}_{k=1}^K$ are supported on discrete points $\mathbf{c} = \{\mathbf{c}_k \in \mathbb{R}^D\}_{k=1}^K$, while atom weights $\pi = (\pi_1, \dots, \pi_K)^\top$, with $\pi_k \geq \epsilon_\pi > 0$ and $\sum_{k=1}^K \pi_k = 1$, specify total mass received by each atom to reflect the atom importance.

The multi-view discrete OT map is denoted as $\mathcal{MOT} = \{T^v : \{\mathbf{x}_i^v\}_{i=1}^N \rightarrow \{\mathbf{c}_k\}_{k=1}^K \mid T_{\#}^v \mu^v = \nu\}_{v=1}^V$. We solve for a set of multi-view OT plans $\mathcal{T} = \{T^v \in \Pi(\mu^v, \nu)\}_{v=1}^V$ by minimizing total transport costs from source distributions $\{\mu^v\}_{v=1}^V$ to the target distribution ν ,

$$\begin{aligned} \mathcal{MOT}(\{\mu^v\}_{v=1}^V, \nu) &= \sum_{v=1}^V W_2^2(\mu^v, \nu) \\ &= \min_{\{T^v \in \Pi(\mu^v, \nu)\}_{v=1}^V} \sum_{v=1}^V \langle T^v, \mathbf{M}^v \rangle_F, \end{aligned} \quad (7)$$

where the squared 2-Wasserstein distance $W_2^2(\mu^v, \nu)$ quantifies the minimum transport cost between source instances $\{\mathbf{x}_i^v\}_{i=1}^N$ and target points $\{\mathbf{c}_k\}_{k=1}^K$. $\mathbf{M}^v \in \mathbb{R}^{N \times K}$ is the cost matrix, of which element measures M_{ik}^v transport cost between $\mathbf{x}_i^v$ and $\mathbf{c}_k$. T^v couples $\{\mathbf{x}_i^v\}_{i=1}^N$ with $\{\mathbf{c}_k\}_{k=1}^K$ under the marginal constraints $\Pi(\mu^v, \nu) = \{T^v \in \mathbb{R}_+^{N \times K} \mid T^v \mathbf{1}_N = \mathbf{a}^v, (T^v)^\top \mathbf{1}_K = \pi\}$ for measure preservation.

3.2.2 Multi-view Shared OT Barycenter

The OT Barycenter is defined as the global semantic-geometric center of source distributions $\mathcal{U} = \{\mu^v\}_{v=1}^V$ in the common measure space, parameterized by semantic centers $\{\mathbf{c}_k^*\}_{k=1}^K$ with importance weights $\{\pi_k^*\}_{k=1}^K$ and solved by

$$\beta^* = \arg \min_{\beta \in \mathcal{P}_K} \sum_{v=1}^V W_2^2(\mu^v, \beta), \quad (8)$$

where $\mathcal{P}_K$ is the candidate barycenter set restricted to the K -atomic family

$$\mathcal{P}_K = \left\{ \beta = \sum_{k=1}^K \pi_k \delta_{\mathbf{c}_k} \mid \mathbf{c}_k \in \mathcal{K}, \pi_k \geq \epsilon_\pi, \sum_{k=1}^K \pi_k = 1 \right\}, \quad (9)$$

where $\mathcal{K} \subset \mathbb{R}^D$ is compact and $0 < \epsilon_\pi < 1/K$. The constraint $\pi_k \geq \epsilon_\pi$ ensures that all K barycenter atoms remain active.

- **Adaptive Learning:** Serving as the learnable target distribution in multi-view OT map, the transport barycenter β is solved together with the measure-preserving transport plans $\mathcal{T} = \{T^v \in \Pi(\mu^v, \beta)\}_{v=1}^V$, by minimizing the total transport cost in Eq. (7). Thus, β is adaptively learned from multi-view distributions in a data-driven manner.
- **Global Anchor:** As the minimizer of the total transport cost over $\mathcal{P}_K$,

$$\mathcal{J}(\beta^*) = \min_{\beta \in \mathcal{P}_K} \sum_{v=1}^V \min_{T^v \in \Pi(\mu^v, \beta)} \langle T^v, \mathbf{M}^v \rangle_F, \quad (10)$$

the OT Barycenter $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*}$ serves as a global anchor that self-perceives intrinsic cluster manifolds by aligning and aggregating view-specific semantic-geometric information of $\{\mu^v\}_{v=1}^V$ while transporting unbiased and consistent cluster information for $\{\mu^v\}_{v=1}^V$.

3.3 Advantages of MvDOT

3.3.1 Clustering-Friendly Properties of OT Barycenter

Property 1 (Global Optima Existence) For multi-view source distributions $\{\mu^v\}_{v=1}^V$, there exists at least one OT barycenter $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*} \in \mathcal{P}_K$ that minimizes the total transport cost,

$$\mathcal{J}(\beta^*) = \min_{\beta \in \mathcal{P}_K} \mathcal{J}(\beta), \quad \mathcal{J}(\beta) = \sum_{v=1}^V W_2^2(\mu^v, \beta). \quad (11)$$

Property 2 (Semantic Region Separability) The OT barycenter $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*}$ with distinct support points $\mathbf{c}_k^* \neq \mathbf{c}_l^*$ for all $k \neq l$, induces consensus semantic regions shared by multi-view source distributions $\{\mu^v\}_{v=1}^V$,

$$\mathcal{R}(\mathbf{c}_k^*) = \{\mathbf{x} \in \Omega \mid \langle \mathbf{x}, \mathbf{c}_k^* \rangle + \phi_k \geq \langle \mathbf{x}, \mathbf{c}_l^* \rangle + \phi_l, \forall k \neq l\}, \quad (12)$$

where $\phi_k \in \mathbb{R}$ denotes the dual potential associated with the barycenter support $\mathbf{c}_k^*$. Together, they define the transport regions: $\mathbf{c}_k^*$ determines the normal direction of partition hyperplanes, while ϕ_k controls their boundary location. The common space $\Omega \subset \mathbb{R}^D$ is partitioned into K disjoint convex regions $\{\mathcal{R}(\mathbf{c}_k^*)\}_{k=1}^K$,

$$\Omega = \bigcup_{k=1}^K \mathcal{R}(\mathbf{c}_k^*), \quad \text{int}(\mathcal{R}(\mathbf{c}_k^*)) \cap \text{int}(\mathcal{R}(\mathbf{c}_l^*)) = \emptyset, \quad \forall k \neq l. \quad (13)$$

Proof of Property 1. We prove the existence by verifying condition validity for the Weierstrass extreme value theorem.

(1) **Parameter Space Compactness.** Each $\beta \in \mathcal{P}_K$ is parameterized by $\{\mathbf{c}_k\}_{k=1}^K$ and π , where the support point $\mathbf{c}_k \in \mathcal{K}$ and the weight vector π lies in the probability simplex $\Delta_{K, \epsilon_\pi} = \{\pi \in \mathbb{R}^K \mid \pi_k \geq \epsilon_\pi, \sum_{k=1}^K \pi_k = 1\}$. The barycenter family $\mathcal{P}_K$ admits a parameterization over the joint parameter space $\Theta = \mathcal{K}^K \times \Delta_{K, \epsilon_\pi}$. Thus, the minimization of $\mathcal{J}(\cdot)$ over $\mathcal{P}_K$ is equivalently studied through its parameterized form on Θ .

Since $\mathcal{K} \subset \mathbb{R}^D$ is compact, the support space $\mathcal{K}^K$ is compact. Moreover, Δ_{K,ϵ_π} is a nonempty closed and bounded subset of $\mathbb{R}^K$, hence compact. Therefore, the joint parameter space Θ is compact. Note that restricting the support $\mathbf{c}_k$ to a compact set $\mathcal{K}$ is reasonable: under the squared Euclidean cost, any sequence of centroids with $\|\mathbf{c}_k\| \rightarrow +\infty$ will drive the transport cost $\|\mathbf{x}_i^v - \mathbf{c}_k\|^2 \rightarrow +\infty$, which causes $\mathcal{J}(\beta) \rightarrow +\infty$. This coercivity ensures that any minimizer of $\mathcal{J}(\cdot)$ lies in a bounded region of $\mathbb{R}^d$.

(2) Objective Function Continuity. Define the transport cost $\mathbf{M}_\beta^v \in \mathbb{R}^{N \times K}$, with entries $[\mathbf{M}_\beta^v]_{ik} = \|\mathbf{x}_i^v - \mathbf{c}_k\|^2$. Since the squared Euclidean distance is continuous in $\mathbf{c}_k$, $\mathbf{M}_\beta^v$ depends continuously on the support parameters $\{\mathbf{c}_k\}_{k=1}^K$.

The squared Wasserstein distance is given by the OT cost:

$$W_2^2(\mu^v, \beta) = \min_{\mathbf{T}^v \in \Pi(\mu^v, \beta)} \langle \mathbf{T}^v, \mathbf{M}_\beta^v \rangle_F, \quad (14)$$

where the feasible set of couplings is

$$\Pi(\mu^v, \beta) = \left\{ \mathbf{T}^v \in \mathbb{R}_+^{N \times K} \mid \mathbf{T}^v \mathbf{1}_K = \mathbf{a}^v, (\mathbf{T}^v)^\top \mathbf{1}_N = \boldsymbol{\pi} \right\}. \quad (15)$$

For each fixed β , $\Pi(\mu^v, \beta)$ is a nonempty compact polytope that varies continuously with β . Since the objective $\langle \mathbf{T}^v, \mathbf{M}_\beta^v \rangle$ is continuous in both $\mathbf{T}^v$ and β , the value function $\beta \mapsto W_2^2(\mu^v, \beta)$ is continuous by Berge's Maximum Theorem. Since the total objective $\mathcal{J}(\beta)$ is a finite sum of continuous functions, it is also continuous on $\mathcal{P}_K$.

(3) Transport Minimizer Existence. Because $\mathcal{P}_K$ is parameterized by the compact set Θ , and $\mathcal{J}$ is continuous on $\mathcal{P}_K$, the Weierstrass extreme value theorem implies that $\mathcal{J}(\cdot)$ attains its minimum on $\mathcal{P}_K$. Hence there exists at least one global minimizer

$$\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*} \in \mathcal{P}_K, \text{ s.t. } \mathcal{J}(\beta^*) = \min_{\beta \in \mathcal{P}_K} \mathcal{J}(\beta). \quad (16)$$

Proof of Property 2. We prove by contradiction that the semantic regions induced by the OT barycenter are pairwise disjoint.

(1) Space Coverage. For any $\mathbf{x} \in \Omega$, the finite set $\{\langle \mathbf{x}, \mathbf{c}_k^* \rangle + \phi_k\}_{k=1}^K$ admits at least one maximizer. Hence, there exists some $k \in \{1, \dots, K\}$ such that

$$\langle \mathbf{x}, \mathbf{c}_k^* \rangle + \phi_k \geq \langle \mathbf{x}, \mathbf{c}_l^* \rangle + \phi_l, \quad \forall k \neq l, \quad (17)$$

which implies $\mathbf{x} \in \mathcal{R}(\mathbf{c}_k^*)$, and $\Omega = \bigcup_{k=1}^K \mathcal{R}(\mathbf{c}_k^*)$.

(2) Region Disjointness and Convexity. Assume there exist $k \neq l$ and a point $\mathbf{x}' \in \text{int}(\mathcal{R}(\mathbf{c}_k^*)) \cap \text{int}(\mathcal{R}(\mathbf{c}_l^*))$. By the definition of interior, there exists an open neighborhood $U(\mathbf{x}') \subset \Omega$ such that $U(\mathbf{x}') \subset \mathcal{R}(\mathbf{c}_k^*) \cap \mathcal{R}(\mathbf{c}_l^*)$.

Hence, for every $\mathbf{y} \in U(\mathbf{x}')$,

$$\begin{aligned} \langle \mathbf{y}, \mathbf{c}_k^* \rangle + \phi_k &\geq \langle \mathbf{y}, \mathbf{c}_l^* \rangle + \phi_l, \\ \langle \mathbf{y}, \mathbf{c}_l^* \rangle + \phi_l &\geq \langle \mathbf{y}, \mathbf{c}_k^* \rangle + \phi_k. \end{aligned} \quad (18)$$

Therefore,

$$\langle \mathbf{y}, \mathbf{c}_k^* \rangle + \phi_k = \langle \mathbf{y}, \mathbf{c}_l^* \rangle + \phi_l, \quad \forall \mathbf{y} \in U(\mathbf{x}'). \quad (19)$$

which means $U(\mathbf{x}')$ is contained in the decision boundary

$$\partial_{kl} = \{\mathbf{x} \in \Omega \mid \langle \mathbf{x}, \mathbf{c}_k^* \rangle + \phi_k = \langle \mathbf{x}, \mathbf{c}_l^* \rangle + \phi_l\}. \quad (20)$$

Expanding the above equality Eq. (20) gives $\langle \mathbf{x}, \mathbf{c}_k^* - \mathbf{c}_l^* \rangle + \phi_k - \phi_l = 0$, which defines an affine hyperplane in $\mathbb{R}^D$. Since $\mathbf{c}_k^* \neq \mathbf{c}_l^*$, this hyperplane has codimension one and thus empty interior in Ω . This contradicts the existence of the open neighborhood $U(\mathbf{x}')$. Hence, $\text{int}(\mathcal{R}(\mathbf{c}_k^*)) \cap \text{int}(\mathcal{R}(\mathbf{c}_l^*)) = \emptyset$, $\forall k \neq l$. Moreover, each $\mathcal{R}(\mathbf{c}_k^*)$ is convex, since it is the intersection of half-spaces induced by affine inequalities Eq. (17).

3.3.2 MvDOT for Sample-to-Cluster Transport Matching

Theorem 1. Given a dataset $\mathcal{X}$ suffering from semantic- and instance-level misalignment as stated in Defs. 1 and 2, MvDOT learns the OT barycenter $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*}$ to construct consensus semantic regions $\{\mathcal{R}(\mathbf{c}_k^*)\}_{k=1}^K$ guaranteed by Prop. 1 and 2, and adopts the sample-to-cluster map $\{T^v : \{\mathbf{x}_i^v\}_{i=1}^N \rightarrow \{\mathbf{c}_k^*\}_{k=1}^K \mid T_\#^v \mu^v = \beta^*\}_{v=1}^V$ to transport samples $\{\mathbf{x}_i^v\}_{i=1}^N$ toward consensus regions $\{\mathcal{R}(\mathbf{c}_k^*)\}_{k=1}^K$ anchored by $\{\mathbf{c}_k^*\}_{k=1}^K$.

Under the guidance of β^* , this process is governed by both semantic consistency and measure equivalence constraints, to form stable cluster structures while preserving discriminative semantic boundaries. To be specific,

- (i) **Semantic Consistency:** View-specific manifold $\mathcal{S}_k^v$ is uniformly transport to the k -th center $\mathbf{c}_k^*$, satisfying $T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*$.
- (ii) **Measure Equivalence:** For view v , the pushforward $T_\#^v \mu^v = \beta^*$ ensures total transported mass assigned to $\mathbf{c}_k^*$ is identical across views, satisfying $\mu^v((T^v)^{-1}(\mathbf{c}_k^*)) = \pi_k^* \Rightarrow \mu^v(\mathcal{S}_k^v) = \pi_k^*$.

Thus, cross-view discrete manifolds with consistent semantics and equivalent measures $\{\mathcal{S}_k^v\}_{v=1}^V$ are aggregated within the consensus region $\mathcal{R}(\mathbf{c}_k^*)$ with continuous convex structure, i.e., $\{T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*, \mu^v(\mathcal{S}_k^v) = \pi_k^*\}_{v=1}^V \Rightarrow \{\mathcal{S}_k^v\}_{v=1}^V \rightarrow \mathcal{R}(\mathbf{c}_k^*)$.

Proof: Consider the empirical sample size $N \rightarrow \infty$, the discrete measure μ^v weakly converges to an absolutely continuous measure $\tilde{\mu}^v$ ($d\tilde{\mu}^v(x) = \rho^v(x)dx$). For the quadratic-cost semi-discrete OT problem from continuous measure $\tilde{\mu}^v$ to the discrete measure β^* , Brenier's theorem ensures the existence of a unique convex function $u^v : \mathbb{R}^d \rightarrow \mathbb{R}$ to induce an OT map T^v as the gradient almost everywhere, i.e., $T^v(x) = \nabla u^v(x)$.

(1) Semantic Assignment Transport. Due to the target measure $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*}$ supported on a finite set $\{\mathbf{c}_k^* \in \mathbb{R}^D\}_{k=1}^K$, the convex potential u^v for the semi-discrete OT problem can be represented as the pointwise maximum of affine functions: $u^v(\mathbf{x}^v) = \max_{k=1}^K \{\langle \mathbf{c}_k^*, \mathbf{x}^v \rangle + h_k^v\}$, where $\langle \mathbf{c}_k^*, \mathbf{x}^v \rangle$ measures the semantic correlation between source instance $\mathbf{x}^v$ and target center $\mathbf{c}_k^*$, and the intercept $h_k^v \in \mathbb{R}$ determines the decision boundary.

- Singularity partitioning: The non-differentiable points of $u^v(\mathbf{x}^v)$ form decision boundaries,

$$\partial_{kl}^v = \{\mathbf{x}^v \mid \langle \mathbf{c}_k^*, \mathbf{x}^v \rangle + h_k^v = \langle \mathbf{c}_l^*, \mathbf{x}^v \rangle + h_l^v, \forall k \neq l\}, \quad (21)$$

where the singularity set ∂_{kl}^v allows the transport map T^v to jump at discontinuities of semantic distribution, rather enforcing continuous interpolation, which maintains the natural semantic boundary of each view.

- Cell decomposition: Based on Property 2, we instantiate the consensus regions $\{\mathcal{R}(\mathbf{c}_k^*)\}_{k=1}^K$ as view-specific transport cells $\{\mathcal{W}_k^v\}_{k=1}^K$ that are induced by their potential u^v to partition the view-specific space Ω^v into K mutually exclusive convex regions:

$$\begin{aligned} \Omega &= \bigcup_{k=1}^K \mathcal{W}_k^v, \text{ int}(\mathcal{W}_k^v) \cap \text{int}(\mathcal{W}_l^v) = \emptyset, \quad \forall k \neq l, \\ \mathcal{W}_k^v &= \{\mathbf{x}^v \mid \langle \mathbf{c}_k^*, \mathbf{x}^v \rangle + h_k^v \geq \langle \mathbf{c}_l^*, \mathbf{x}^v \rangle + h_l^v, \forall k \neq l\}, \end{aligned} \quad (22)$$

Where the induced OT map T^v is constant on each cells $\mathcal{W}_k^v$, i.e., $T^v(\mathcal{W}_k^v) = \mathbf{c}_k^*$. Geometrically, the intercept h_k^v adaptively adjusts the transport cells $\mathcal{W}_k^v$ to compensate for view-specific distributional deviations between the source distribution μ^v and the OT barycenter β^* , which

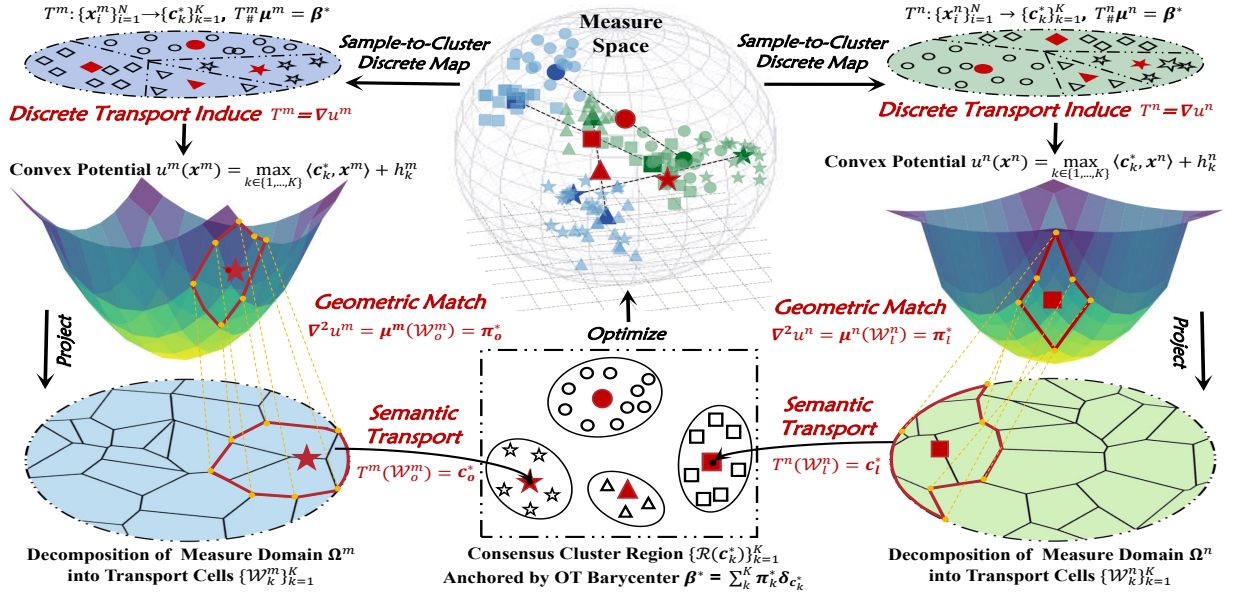


Fig. 2: Sample-to-Cluster Discrete Mapping. By aggregating semantic-geometric information of multi-view distributions $\mathcal{U}$, the OT barycenter β^* is learned as a target distribution to align and fuse cross-view discrete cluster manifolds. **Transport Induce:** To transport $\{\mu^v\}_{v=1}^V$ to $\beta^* = \sum_{k=1}^K \pi_k^* \delta_{\mathbf{c}_k^*}$, MvDOT induces discrete OT maps $\{T^v = \nabla u^v\}_{v=1}^V$ via convex potentials $u^v(\mathbf{x}^v) = \max_{k \in \{1, \dots, K\}} \{\langle \mathbf{c}_k^*, \mathbf{x}^v \rangle + h_k^v\}$. As the maximum over K affine functions, u^v partitions view-specific space Ω^v into K disjoint transport cells $\{\mathcal{W}_k^v\}_{k=1}^K$ with $T^v(\mathcal{W}_k^v) = \mathbf{c}_k^*$, where view-specific manifold $\mathcal{S}_k^v$ semantically consistent with $\mathcal{W}_k^v$ is also transported to the same center $\mathbf{c}_k^*$, i.e. $T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*$. **Semantic Transport:** Within each $\mathcal{S}_k^v$, samples $\mathbf{x}^v \in \mathcal{S}_k^v$ are jointly transported in many-to-one manner. **Geometric Match:** The push-forward constraint ensures that each view allocates the transport mass $\mu^v(\mathcal{W}_k^v)$ equal to the sample scale π_k^* of the consensus region $\mathcal{R}(\mathbf{c}_k^*)$ anchored by $\mathbf{c}_k^*$. Cross-view manifolds $\{\mathcal{S}_k^v\}_{v=1}^V$ with similar semantics $T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*$ and same measure $\mu^v(\mathcal{S}_k^v) = \pi_k^*$ are uniformly integrated into $\mathcal{R}(\mathbf{c}_k^*)$.

ensure instances belonging to the same semantic manifold $\mathcal{S}_k^v$ fall into their matched cell $\mathcal{W}_k^v$, i.e., $\mathcal{S}_k^v \subseteq \mathcal{W}_k^v$.

Since both view-shared semantic regions $\mathcal{R}(\mathbf{c}_k^*)$ and view-specific transport cells $\mathcal{W}_k^v$ are guided by the semantic centers $\{\mathbf{c}_k^*\}_{k=1}^K$, $\mathcal{W}_k^v$ serves as view-adapted realization of $\mathcal{R}(\mathbf{c}_k^*)$ via the OT map T^v , to achieve cross-view semantic alignment at the level of shared transport destinations with consistent cluster indexing. More specifically, for $\mathbf{x}^v \in \mathcal{S}_k^v \subseteq \mathcal{W}_k^v$, it follows

$$T^v(\mathcal{W}_k^v) = \mathbf{c}_k^* \Rightarrow T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*. \quad (23)$$

(2) Geometric Measure Matching: For pushing each source measure μ^v to the OT barycenter β^* , the map T^v satisfies the pushforward condition $T^v_{\#} \mu^v = \beta^*$. Since β^* is a discrete measure, the pushforward condition is equivalent to the mass conservation constraints,

$$\mu^v((T^v)^{-1}(\mathbf{c}_k^*)) = \pi_k^*, \forall k \in \{1, \dots, K\}. \quad (24)$$

According to cell decomposition, since semantically similar samples inside the cell, i.e., $\mathbf{x}^v \in \mathcal{S}_k^v \subseteq \mathcal{W}_k^v$ share the same transport destination $\mathbf{c}_k^*$, the cell $\mathcal{W}_k^v$ is mapped to the center $\mathbf{c}_k^*$ (up to a set of μ^v -measure zero):

$$(T^v)^{-1}(\mathbf{c}_k^*) = \{\mathbf{x}^v \in \Omega^v \mid T^v(\mathbf{x}^v) = \mathbf{c}_k^*\} \equiv \mathcal{W}_k^v. \quad (25)$$

Since $\mathcal{S}_k^v = \mathcal{W}_k^v$ (μ^v -a.e.), substituting the identity in Eq. 25 into Eq. 24 yields,

$$\mu^v(\mathcal{W}_k^v) = \pi_k^* \Rightarrow \mu^v(\mathcal{S}_k^v) = \pi_k^*, \forall k \in \{1, \dots, K\}. \quad (26)$$

Under joint semantic-geometric constraints $\{T^v(\mathcal{S}_k^v) = \mathbf{c}_k^*, \mu^v(\mathcal{S}_k^v) = \pi_k^*\}_{v=1}^V$, a sample-to-cluster transport mapping is induced to promote cross-view consistency matching between discrete cluster manifolds $\{\mathcal{S}_k^v\}_{v=1}^V$. These manifolds, semantically

similar to the semantic center $\mathbf{c}_k^*$ with identical source mass π_k^* assigned to $\mathbf{c}_k^*$, are merged into the semantically and geometrically coherent convex region $\mathcal{R}(\mathbf{c}_k^*)$, i.e., $\{\mathcal{S}_k^v\}_{v=1}^V \rightarrow \mathcal{R}(\mathbf{c}_k^*)$.

4 INSTANTIATION OF MvDOT FOR UMVC TASK

In this section, MvDOT is instantiated as a cluster-level semantic-structure matching framework in Fig. 3, to handle dual-unaligned challenges caused by discrete manifolds and unpaired samples.

4.1 Representation Reconstruction

To ensure semantic separability and structural stability for matching and clustering, we adopt independent autoencoders per view to uncover the intrinsic low-dimensional manifold of high-dimensional data. Each autoencoder $\{\mathcal{E}^v, \mathcal{D}^v\}_{v=1}^V$ embeds view data $\mathbf{X}^v$ into latent representations $\mathbf{Z}^v$ in a view-specific latent space $\Omega^v \subset \mathbb{R}^D$ and recovers them via the decoder by minimizing the reconstruction loss:

$$\begin{aligned} \mathcal{L}_{rec} &= \sum_{v=1}^V \|\mathbf{X}^v - \mathcal{D}^v(\mathbf{Z}^v)\|_2^2 \\ &= \sum_{v=1}^V \sum_{i=1}^{N_v} \|\mathbf{x}_i^v - \mathcal{D}^v(\mathcal{E}^v(\mathbf{x}_i^v))\|_2^2, \end{aligned} \quad (27)$$

where $\mathcal{E}^v(\mathbf{X}^v; \theta^v) : \mathbf{X}^v \in \mathbb{R}^{N \times d_v} \rightarrow \mathbf{Z}^v \in \mathbb{R}^{N \times D}$ and $\mathcal{D}^v(\mathbf{Z}^v; \varphi^v) : \mathbf{Z}^v \in \mathbb{R}^{N \times D} \rightarrow \hat{\mathbf{X}}^v \in \mathbb{R}^{N \times d_v}$.

4.2 Discrete Manifold Transport

Based on Thm. 1, we instantiate MvDOT as a barycenter-guided sample-to-cluster transport process to integrate discrete manifolds

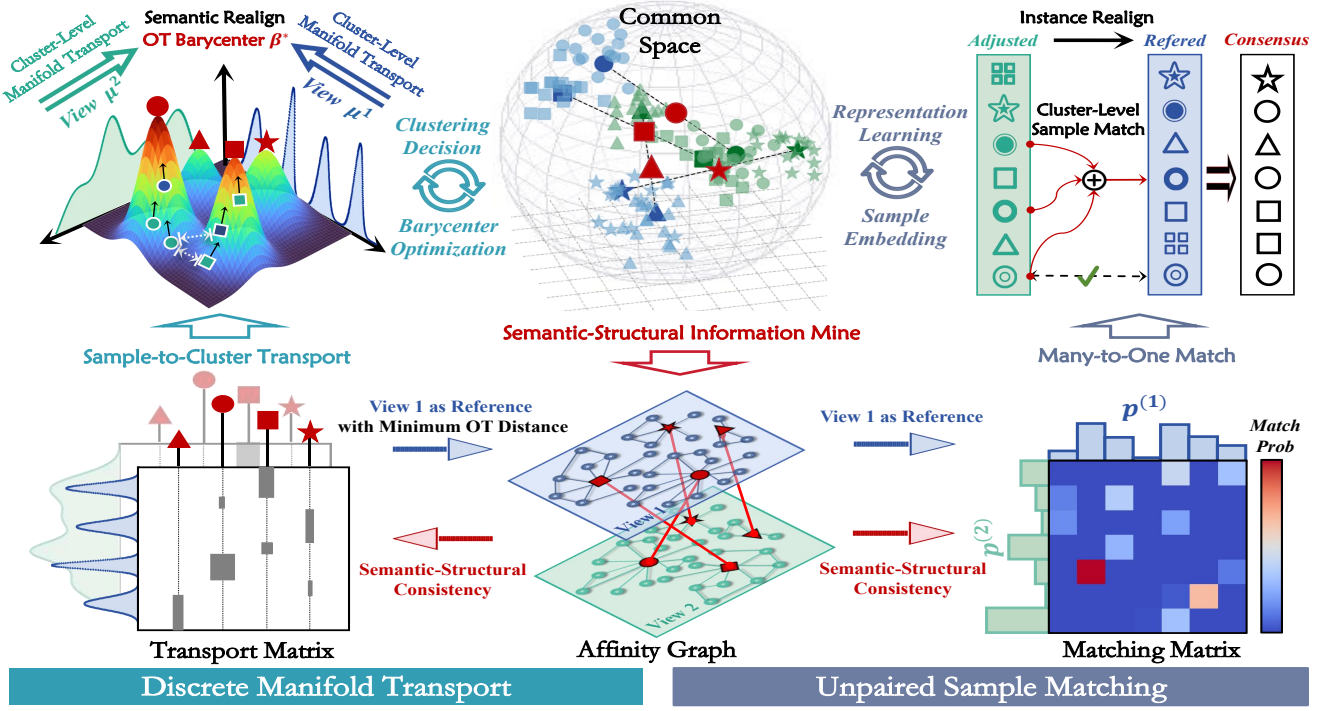


Fig. 3: Instantiation of MvDOT. First, multi-view data are embedded into common space and the transport barycenter is initialized via K-means; then, MvDOT minimizes total transport cost between all samples and the barycenter to learn an OT barycenter and mine barycenter-anchored consensus clusters for cross-view alignment and clustering. **(a) Discrete manifold transport:** Under the guidance of the OT barycenter, discrete manifolds are transported to consensus clusters via consistency constraint on semantic membership and geometric metric. Via sample-to-cluster matching, MvDOT integrates discrete manifolds with similar semantics and structures into a consensus cluster manifold to achieve cluster-level consistency across view. **(b) Unpaired sample matching:** Based on the OT barycenter, cross-view affinity graphs are constructed to unbiasedly characterize semantic distributions and topological structures and cross-view semantic-topological discrepancies are optimized to globally seek reliable sample correspondences. Via many-to-one matching, MvDOT recovers cluster-level correspondences across views.

into continuous convex structures within each cluster and maintain discrete non-convex structures across clusters. Specifically, MvDOT learns an OT barycenter as transport bridge, and guides discrete manifolds toward barycenter-induced consensus clusters under joint semantic-geometric consistency constraints. This process preserves intrinsic semantic boundaries while alleviating intra-cluster manifold dispersion and inter-cluster spurious connections caused by semantic- and instance-level unaligned noise.

4.2.1 Transport Barycenter Initialization

As defined in Sec. 3.2.2, the transport barycenter β is composed of semantic centers $\mathbf{c} = \{\mathbf{c}_k \in \mathbb{R}^D\}_{k=1}^K$ and importance weights $\pi = \{\pi_k \in \mathbb{R}\}_{k=1}^K$ parameterized as $\mathcal{B} = \{(\mathbf{c}_k, \pi_k)\}_{k=1}^K$, and its mixture distribution is further expressed as $\beta = \sum_{k=1}^K \pi_k \delta_{\mathbf{c}_k}$. We stack all view latent representations $\mathcal{Z} = \{\mathbf{Z}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$ row-wise to build a full-information matrix $\mathbf{Z} = [\mathbf{Z}^1; \dots; \mathbf{Z}^V] \in \mathbb{R}^{(NV) \times D}$, and employ spherical K-means to initialize each center $\mathbf{c}_k$ and assign a uniform semantic weight $\pi_k = 1/K$ for all k :

$$\min_{\{\mathbf{c}_k\}_{k=1}^K} \sum_{i=1}^{NV} \sum_{k=1}^K \|\mathbf{z}_i - \mathbf{c}_k\|_2^2, \quad \text{s.t. } \|\mathbf{c}_k\|_2 = 1, \forall k, \quad (28)$$

where unit-norm constraint $\|\mathbf{c}_k\|_2 = 1$ stabilizes center directions to enhance geometric discrimination and encourages compact and well-separated clusters with mutually exclusive semantics.

4.2.2 Semantic-geometric Transport Matching

MvDOT models a barycenter-guided discrete transport mapping from samples to clusters, and minimizes multi-view transport

cost to promote semantic-geometric consistent matching among discrete cluster manifolds. Specifically, give multi-view source measures $\mathcal{U} = \{\mu^v = \sum_{i=1}^N a_i^v \delta_{\mathbf{x}_i^v}\}_{v=1}^V$, MvDOT adaptively learns a transport barycenter $\beta = \sum_{k=1}^K \pi_k \delta_{\mathbf{c}_k}$ as their shared target measure. The Multi-view OT map $\mathcal{MOT}$ in Eq. 29 is implemented as a view-shared transport matching network $\mathcal{F}_m(\mathcal{Z}; \omega)$, to map view-specific representations $\mathcal{Z} = \{\mathbf{Z}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$ from their respective latent spaces $\{\Omega^v\}_{v=1}^V$ into consensus representations $\mathcal{H} = \{\mathbf{H}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$ in a common space Ω . For each view, compute their cost matrix $\mathbf{M}^v \in \mathbb{R}^{N \times K}$ with entries $\mathbf{M}_{ik}^v = \|\mathbf{h}_i^v - \mathbf{c}_k\|_2^2$. By minimizing multi-view transport cost to solve for coupled OT plans $\mathcal{T} = \{\mathbf{T}^v \in \Pi(\mu^v, \beta)\}_{v=1}^V$:

$$\begin{aligned} \mathcal{L}_{mot} &= \min_{\{\mathbf{T}^v \in \mathbb{R}_+^{N \times K}\}_{v=1}^V} \sum_{v=1}^V \langle \mathbf{T}^v, \mathbf{M}^v \rangle_F, \\ \text{s.t. } &\mathbf{T}^v \mathbf{1}_K = \mathbf{a}^v, \quad (\mathbf{T}^v)^\top \mathbf{1}_N = \pi, \end{aligned} \quad (29)$$

where $\mathbf{T}^v$ assigns the sample mass $\mathbf{a}^v = (a_1^v, \dots, a_N^v)^\top$ from view instances $\mathbf{H}^v = \{\mathbf{h}_i^v\}_{i=1}^N$ to center atoms $\mathbf{c} = \{\mathbf{c}_k\}_{k=1}^K$, according to the semantic importance $\pi = (\pi_1, \dots, \pi_K)^\top$.

Considering that consistency constraints tends to disrupt within-view neighborhood relationships and intra-cluster distribution patterns, they may hinder the recovery of intrinsic manifold structure[58, 59]. To preserve topological information, we construct view-specific neighborhood graph $\mathbf{S}^v$ based on view-specific representation $\mathbf{Z}^v$ and impose a Laplacian regularizer

$\text{Tr}((\mathbf{Z}^v)^\top \mathbf{L}^v \mathbf{Z}^v)$ upon $\mathcal{L}_{mot}$:

$$\mathcal{L}_{mm} = \sum_{v=1}^V \langle \mathbf{T}^v, \mathbf{M}^v \rangle_F + \lambda \text{Tr}((\mathbf{Z}^v)^\top \mathbf{L}^v \mathbf{Z}^v), \quad (30)$$

where $\lambda > 0$ balances the learning of view-shared semantics and the preservation of view-specific structures. Integrating local structural alignment into global semantic consistency, this regularizer enhances the semantic similarity of adjacent samples to strengthen intra-cluster compactness and enlarges the semantic divergence of non-adjacent samples to ensure inter-cluster separability. It preserves non-convex data structures with clear cluster boundaries from being corrupted by unreliable consistency alignment.

4.2.3 Transport Barycenter Optimization

To evolve the initial barycenter β into the global optimal center β^* , we separately update centers $\mathbf{c} = \{\mathbf{c}_k\}_{k=1}^K$ and weights $\pi = \{\pi_k\}_{k=1}^K$ online during transport-plan optimization with an entropic Sinkhorn solver. Firstly, the centers $\mathbf{c}$ are parameterized by a learnable network $\mathcal{F}_b(\mathcal{Z}, \mathcal{T}; \psi)$. Fixed transport couplings $\mathcal{T} = \{\mathbf{T}^v\}_{v=1}^V$, the transport objective $\mathcal{L}_{mot}$ admits closed-form gradient to refine each center $\mathbf{c}_k$:

$$\frac{\partial \mathcal{L}_{mot}}{\partial \mathbf{c}_k} = 2 \sum_{v=1}^V \sum_{i=1}^N \mathbf{T}_{ik}^v (\mathbf{c}_k - \mathbf{h}_i^v) = 2 \tau_k (\mathbf{c}_k - \mathbf{s}_k), \quad (31)$$

$$\text{where } \tau_k = \sum_{v=1}^V \sum_{i=1}^N \mathbf{T}_{ik}^v, \quad \mathbf{s}_k = \frac{1}{\tau_k} \sum_{v=1}^V \sum_{i=1}^N \mathbf{T}_{ik}^v \mathbf{h}_i^v.$$

When set the stationary condition $\nabla_{\mathbf{c}_k} \mathcal{L}_{mot} = 0$ gives $\mathbf{c}_k = \mathbf{s}_k$, each center equals the transport-weighted centroid of its assigned samples to accurately capture global semantics. With differentiable Sinkhorn OT, update $\mathbf{c}_k$ by gradient descent and project it to unit norm with numerical stability for spherical centers:

$$\mathbf{c}_k \leftarrow \mathbf{c}_k - \eta \frac{\partial \mathcal{L}_{mot}}{\partial \mathbf{c}_k}, \quad \mathbf{c}_k \leftarrow \frac{\mathbf{c}_k}{\|\mathbf{c}_k\|_2}. \quad (32)$$

Subsequently, the weights π are momentum-smoothed with assignment masses aggregated over the transport couplings $\mathcal{T}$, which encourages the prior uniform weights to follow empirical preferences. For N samples, define cross-view aggregated assignment mass as $\mathbf{T}_{ik} = \sum_{v=1}^V \mathbf{T}_{ik}^v$, the empirical average assignment is $\hat{\pi}_k = \frac{1}{N} \sum_{i=1}^N \mathbf{T}_{ik}$. Update π :

$$\pi_k \leftarrow \alpha \pi_k + (1 - \alpha) \hat{\pi}_k, \quad k \in \{1, \dots, K\}, \quad (33)$$

where $\alpha \in [0, 1)$ is the learning momentum and $\sum_{k=1}^K \hat{\pi}_k = 1$ ensures $\sum_{k=1}^K \pi_k = 1$ after update. The updated π is used as the target marginal in subsequent Sinkhorn iterations.

4.3 Unpaired Sample Matching

We instantiate MvDOT as barycenter-driven many-to-one matching scheme to recover cluster-level sample correspondence with global consistent semantics and local isomorphic topologies. Specifically, as an alignment anchor, the OT barycenter constructs affinity graphs to encode both intra- and inter-view topological relations under unbiased semantic guidance. By incorporating neighborhood relations with semantic information, MvDOT constructs a cluster-aware “detector” to explore reliable samples correspondences and formalizes this exploration as a semantic-topology transport problem to derive an optimal matching matrix. This mechanism resists pseudo-semantic interference induced by unaligned noise and provides more adaptive and robust cluster-level correspondences compared with instance-level counterparts.

4.3.1 Affinity Graph Construction

Given the barycenter $\mathcal{B} = \{(\mathbf{c}_k, \pi_k)\}_{k=1}^K$ as semantic anchors, build intra- and inter-view topological connections in two steps:

- Intra-view anchor graph: For view v , define an anchor graph $\mathbf{G}^v \in \mathbb{R}^{N \times K}$, where each entry $\mathbf{G}_{ik}^v = \exp\left(-\frac{\|\mathbf{h}_i^v - \mathbf{c}_k\|_2^2}{2\sigma^2}\right)$ quantifies the membership strength of i -th instance $\mathbf{h}_i^v$ to k -th center $\mathbf{c}_k$ via Gaussian affinity.
- Inter-view affinity graph: For view m and n , construct the affinity graph as $\mathbf{G}^{mn} = \mathbf{G}^m (\mathbf{G}^n)^\top \in \mathbb{R}^{N \times N}$, which indirectly reflects the correlation strength between cross-view instances via the shared barycenter $\mathcal{B}$.

This construction implicitly propagates global semantic-structural information across views via the transport barycenter, thereby guiding unpaired samples to establish reliable connections at both semantic and structure levels [60].

4.3.2 Topology-Aware Transport

Traditional metrics (e.g., Euclidean and Cosine distance) focus on global geometric gaps among samples in high-dimensional feature space, yet overlook the local topological structure of underlying manifolds in low-dimensional latent space, which yields unreliable similarity measure for heterogeneous manifolds. Topological distance exploits sample correlation to mine neighborhood structure information. Thus, we integrate local topological connectivity with global semantic similarity to design a robust similarity detector.

We discretize the continuous affinity graph $\mathbf{G}^{mn}$ via Breadth-First Search to obtain a topological structure graph Ψ^{mn} where semantic affinity information are decoupled. Only topological connectivity is retained to construct the topological distance $\mathbf{D}_{\mathcal{T}}^{mn} = \sigma(\Psi^{mn})$, where $\Psi_{ij}^{mn} = \text{hops}(\mathbf{G}_{i \rightarrow j}^{mn})$ denotes the shortest-path hops from sample i in view m to sample j in view n , and $\sigma(\cdot)$ denotes sigmoid function that maps unbounded and discrete hops into bounded and smooth structural penalties to make the topological cost more stable and controllable. Given the semantic cost $\mathbf{D}_S^{mn}$ to measure semantic discrepancy, we define the hybrid cost $\mathbf{C}^{mn} = \mathbf{D}_S^{mn} \circ \mathbf{D}_{\mathcal{T}}^{mn}$, which penalizes candidate pairs with low semantic affinity or weak topological connectivity. Based on $\mathbf{C}^{mn}$, sample matching is formulated as an entropy-regularized OT problem to solve the matching matrix $\mathbf{P}^{mn}$:

$$\mathbf{P}^{mn} = \arg \min_{\mathbf{P}^{mn} \in \mathbb{U}(\mathbf{a}^m, \mathbf{a}^n)} \langle \mathbf{P}^{mn}, \mathbf{C}^{mn} \rangle_F + \epsilon \Omega(\mathbf{P}^{mn}), \quad (34)$$

where $\mathbb{U}(\mathbf{a}^m, \mathbf{a}^n)$ is a set of admissible couplings with marginals $\mathbf{a}^m = \mathbf{a}^n = \frac{1}{N} \mathbf{1}_N$, and $\Omega(\mathbf{P}^{mn}) = -\sum_{ij} \mathbf{P}_{ij}^{mn} \log \mathbf{P}_{ij}^{mn}$ is an entropy regularizer solved efficiently via the Sinkhorn-Knopp algorithm with the regularization coefficient ϵ . $\mathbf{P}^{mn}$, a soft doubly stochastic matching, assign higher match priority to candidate pairs with consistent semantics and isomorphic topology.

4.3.3 Semantic-Guided Matching

We introduce a cluster-level matching loss $\mathcal{L}_{sm}$ to dynamically refine the matching matrix $\mathbf{P}^{mn}$ by re-weighting matched pairs according to clustering assignment consistency. Specifically, we maximize the semantic agreement of matched pairs to promote alignment between their cluster-level posterior distributions:

$$\mathcal{L}_{sm}^{mn} = \sum_{i,j=1}^{N_u} \mathbf{P}_{ij}^{mn} \log \left(\sum_{k=1}^K P(l_i^m = k | \mathbf{h}_i^m) P(l_j^n = k | \mathbf{h}_j^n) \right), \quad (35)$$

where $P(l_i^v = k | \mathbf{h}_i^v)$ denotes the posterior probability that the i -th instance $\mathbf{h}_i^v$ belongs to the k -th the centroid $\mathbf{c}_k$.

Without paired priors as supervision signals, the optimization of $\mathcal{L}_{sm}$ may be dominated by views with high-entropy posteriors. Views with ambiguous semantics or loose structures tend to produce highly uncertain posteriors. Such posteriors induce optimization-friendly yet information-scarce losses and mislead model into learning low-confidence information. Conversely, reliable semantic cues from high-quality views is suppressed, weakening discriminative semantics. Thus, we employ a view-weighting scheme to adaptively modulate the information contribution of each view during learning. High-confidence views with smaller deviations from the OT barycenter receive larger weights, whereas low-confidence views with distorted semantics and structures are down-weighted. To support more than two views and enhance cross-view collaboration, we extend $\mathcal{L}_{sm}^{mn}$ to the weighted form:

$$\mathcal{L}_{sm} = \sum_{m=1}^V \sum_{\substack{n=1 \\ n \neq m}}^V \mathbf{W}^m \mathbf{W}^n \mathcal{L}_{sm}^{mn}, \quad (36)$$

where $\mathbf{W}^v$ measures the information reliability of view v via its OT distance to the barycenter, and is normalized among full views:

$$\mathbf{W}^v = \frac{(W_2^2(\boldsymbol{\mu}^v, \boldsymbol{\beta}))^{-1}}{\sum_{v'=1}^V (W_2^2(\boldsymbol{\mu}^{v'}, \boldsymbol{\beta}))^{-1}}, \quad (37)$$

where the OT distance $W_2^2(\boldsymbol{\mu}^v, \boldsymbol{\beta})$ is solved via Eq. 29, and ϵ is a small constant for numerical stability.

Notably, the model adaptively selects a view with the smallest OT distance $v^* = \arg \min_{v \in \{m, n\}} W_2^2(\boldsymbol{\mu}^v, \boldsymbol{\beta})$ as the match reference. Under the guarantee of global consistency alignment across all views and the guidance of the reference view with well-separated semantic structures, this matching mechanism can reliably recover cluster-wise sample correspondences.

4.4 MvDOT for Consistency Matching and Clustering

MvDOT jointly optimizes three complementary tasks: within-view representation reconstruction, cross-view discrete manifold and unpaired sample matching:

$$\mathcal{L} = \mathcal{L}_{rec} + \mathcal{L}_{mm} + \mathcal{L}_{sm}. \quad (38)$$

In practice, MvDOT is implemented as an end-to-end joint optimization framework composed of a differentiable Sinkhorn OT solver and two learnable parametric networks. The OT solver derives the transport plans $\mathcal{T}$ based on the view representations $\mathcal{H}$ and the transport barycenter $\boldsymbol{\beta}$, while the multi-view transport matching network $\mathcal{F}_m$ and the OT barycenter learning network $\mathcal{F}_b$ are jointly optimized to learn the consensus representations and the shared barycenter through gradient backpropagation of the overall loss $\mathcal{L}$. Thanks to this online collaboration, the barycenter is no longer the static geometric average, but cluster-aware anchor jointly driven by matching and clustering objectives under OT constraints. Meanwhile, the transport plan, the barycenter, and the representation adapt to and reinforce one another, which is more effective for learning clustering-friendly consensus representations and discovering well-separated semantic structures. More details refer to Algorithm 1.

Algorithm 1: MvDOT for Matching and Clustering

Input: Dataset $\{\mathbf{X}^v \in \mathbb{R}^{N \times d_v}\}_{v=1}^V$ comprising the aligned and unaligned subset $\{\mathbf{A}^v \in \mathbb{R}^{N_a \times d_v}\}_{v=1}^V$ and $\{\mathbf{U}^v \in \mathbb{R}^{N_u \times d_v}\}_{v=1}^V$; Encoder-Decoders $\{\mathcal{E}^v, \mathcal{D}^v\}_{v=1}^V$, Multi-view transport matching network $\mathcal{F}_m$, OT Barycenter learning network $\mathcal{F}_b$; Warm-up epochs e , Fine-tuning epochs E .

// Prepare for Training

for $t = 1$ **to** e **do**

 Warm up $\{\mathcal{E}^v, \mathcal{D}^v\}_{v=1}^V$ on $\{\mathbf{X}^v\}_{v=1}^V$ with Eq. (27).

end for

Initialize the transport barycenter $\mathcal{B} = \{(\mathbf{c}_k, \pi_k)\}_{k=1}^K$ as elaborated in Sec. 4.2.1.

// Learn for Matching

for $t = 1$ **to** E **do**

 // Sample En-Decoding

for $v = 1$ **to** V **do**

 Encode representations $\mathcal{Z} = \{\mathbf{Z}^v | \mathcal{E}^v(\mathbf{X}^v)\}_{v=1}^V$;

end for

 Compute reconstruction loss $\mathcal{L}_{rec}$ with Eq. (27).

 // Discrete Manifold Transport

 Solve OT plans $\{\mathbf{T}^v \in \Pi(\boldsymbol{\mu}^v, \boldsymbol{\beta})\}_{v=1}^V$ with Eq. (29);

for $v = 1$ **to** V **do**

 Encode representations $\mathcal{H} = \{\mathbf{H}^v | \mathcal{F}_m(\mathbf{Z}^v)\}_{v=1}^V$;

end for

 Calculate manifold matching loss $\mathcal{L}_{mm}$ with Eq. (30).

 // Unpaired Sample Matching

for $m, n = 1$ **to** V ($m \neq n$) **do**

 Construct affinity matrix $\mathbf{G}^{mn}$ as elaborated in Sec. 4.3.1;

 Solve matching matrix $\mathbf{P}^{mn}$ with Eq. (34);

end for

 Calculate sample matching loss $\mathcal{L}_{sm}$ with Eq. (36).

 // Multi-Task Joint Optimization

 Calculate the overall loss $\mathcal{L}$ with Eq. (38);

 Optimize the trainable parameters of $\{\mathcal{E}^v, \mathcal{D}^v\}_{v=1}^V$, $\mathcal{F}_m$ and $\mathcal{F}_b$ via gradient descent to minimize $\mathcal{L}$;

 Adjust the transport barycenter $\mathcal{B} = \{(\mathbf{c}_k, \pi_k)\}_{k=1}^K$ from $\mathcal{F}_b$ via Eq. (32) and Eq. (33).

end for

// Infer for Clustering

Encode aligned, unaligned consensus representations

$\{\mathbf{H}_a^v | \mathcal{E}^v : \mathbf{A}^v \rightarrow \mathbf{Z}_a^v, \mathcal{F}_m : \mathbf{Z}_a^v \rightarrow \mathbf{H}_a^v\}_{v=1}^V$,

$\{\mathbf{H}_u^v | \mathcal{E}^v : \mathbf{U}^v \rightarrow \mathbf{Z}_u^v, \mathcal{F}_m : \mathbf{Z}_u^v \rightarrow \mathbf{H}_u^v\}_{v=1}^V$;

Learn the OT barycenter $\boldsymbol{\beta}^*$ with center $\{\mathbf{c}_k^*\}_{k=1}^K$ via $\mathcal{F}_b$;

Select view $v^* = \arg \min_{v \in \{1, \dots, V\}} W_2^2(\boldsymbol{\mu}^v, \boldsymbol{\beta}^*)$ as alignment reference;

for $v = 1$ **to** V ($v \neq v^*$) **do**

 Based on $\{\mathbf{H}_a^v, \mathbf{H}_u^v\}$ and $\{\mathbf{c}_k^*\}_{k=1}^K$, solve matching matrix $\mathbf{P}^{vv^*}$ using Eq. (34);

for $\mathbf{h}_i^{v^*}$ in $\{\mathbf{H}_u^{v^*}\}$ **do**

 Realign counterpart $\tilde{\mathbf{h}}_i^v$ to $\mathbf{h}_i^{v^*}$ through $\tilde{\mathbf{h}}_i^v = \sum_{j=1}^{N_u} \mathbf{P}_{ij}^{vv^*} \mathbf{h}_j^v$.

end for

end for

Concatenate aligned $\{\mathbf{H}_a^v\}_{v=1}^V$ with realigned $\{\tilde{\mathbf{H}}_u^v\}_{v=1}^V$ to form $\{\tilde{\mathbf{H}}^v \in \mathbb{R}^{N \times D}\}_{v=1}^V$, and obtain clustering representations $\mathbf{H} \in \mathbb{R}^{N \times D}$ via view-weighted sum.

Output: $\hat{\mathbf{Y}} = \{\arg \min_{k \in \{1, \dots, K\}} \|\mathbf{h}_i - \mathbf{c}_k^*\|_2\}_{i=1}^N$.

5 EXPERIMENT

5.1 Experimental Setups

5.1.1 Datasets and Metrics

We comprehensively evaluate our method on six diverse datasets using accuracy (ACC), normalized mutual information (NMI), and adjusted Rand index (ARI).

TABLE 1: Statistics of Benchmark Datasets.

Dataset	Sample	Cluster	View Dimension	Characteristic
CUB	600	10	1024/300	Small-scale samples
Caltech-5V	1400	7	1984/512/928/254/40	Dimension discrepancy
NUS-WIDE	9000	10	4096/300	View Heterogeneity
ALOI-100	10800	100	77/13/64/125	Semantic Diversity
Reuters	18758	6	10/10/10/10/10	Cluster-scale Imbalance
NoisyMNIST	70000	10	784/784	Large-scale with Noise

5.1.2 Baselines

Following the discussion in Introduction and Related Work, we compare MvDOT with ten representative SOTA methods, including PVC [41], GWMAC [48], MvCLN [42], SURE [43], SMILE [26], CGCN [39], RMCNC [21], VITAL [29], CANDY [61], and TCLPVC [5]. They cover diverse design choices in view consistency alignment, sample correspondence matching, and model function adaptation, providing a fair and comprehensive evaluation of MvDOT under complex MVC tasks. To further explore the underlying causes of performance differences, we organize all eleven methods in Table 2 according to the following taxonomy:

- **Alignment Mechanism:** How alignment is performed and what entities are aligned, include both the alignment operator (e.g., continuous smoothing or discrete matching) and the alignment granularity (e.g., samples, distributions, structures or clusters).
- **Alignment Criterion:** The measure or constraint to qualify whether cross-view correspondences are consistent, reflects the level (e.g., instance, view or cluster) and form (e.g., representation similarity, distribution or structure consistency, cluster matching) of consistency alignment.
- **Matching Mechanism:** The strategy is adopted to infer cross-view sample relationships without paired priors, e.g., greedy, Hungarian, or OT-based matching.
- **Matching Granularity:** The hierarchy and form at which sample correspondences are modeled (e.g., instance/neighborhood/cluster-level, as well as one-to-one hard or many-to-one soft matching), reflects the degree of matching agreement and stability of matching results.
- **Assumption Dependency:** The degree to which the effectiveness of consistency alignment relies on data assumptions of manifold characteristics and paired priors, reflects the model adaptability to discrete misaligned datasets.
- **View Scalability:** The number of views that synchronously participate in information interaction, alignment, and fusion, reflects the model's information integration capability and attainable consistent degree.
- **Sample Utilization:** The processing manner and utilization extent for multi-view data, reflects the model ability to mine unbiased information and realize unbiased clustering without paired priors.

- **Applicable Scenarios:** The settings of sample correspondences are adapted by methods (e.g., fully/partially view-aligned, or fully view-unaligned).

Since some baselines are not designed for multi-view inputs or fully unaligned samples, we adopt following settings to ensure fair comparison while preserving their original experimental protocols:

- **View Utilization Setting:** Methods employs the maximum number of views it natively supports. For dual-view restricted methods including MvDOT-2V, the first two views are selected as experimental data.
- **Paired Sample Setting:** The sample order of first view is fixed, while other views are randomly permuted according to aligned rate. Methods tailored for partial unaligned setting support full unaligned inputs in implementation but typically degrade without the guidance of paired priors.

5.2 Implementation details

Models are trained on an NVIDIA 3090 GPU using the Adam optimizer in PyTorch 2.1.0. The hyperparameters are set as follows for different datasets: learning rate $lr \in \{1e-4, 3e-4, 5e-4\}$, batch size $b \in \{256, 512, 1024\}$, entropy regularization $\epsilon \in \{0.1, 0.2\}$, graph regularization $\lambda \in \{0.05, 0.1\}$, and momentum $\alpha \in \{0.95, 0.98\}$. For all datasets, 4-layer autoencoders $\{\mathcal{E}^v, \mathcal{D}^v\}_{v=1}^V$ with identical architectures are employed for each view; the multi-view transport matching network $\mathcal{F}_m$ is a single FC layer, while the barycenter learning network $\mathcal{F}_b$ is a learnable parameter matrix in $\mathbb{R}^{K \times D}$.

5.3 Competitiveness of MvDOT

5.3.1 Superior Adaptability to Complex MVC Tasks

Table 3 and Fig. 4 presents comparison results under three typical scenarios: aligned rates of 0.0, 0.5, and 1.0. On diverse datasets, MvDOT performs strong multi-scenario adaptability on superior effectiveness and robustness.

Effectiveness. MvDOT achieves the best or near-best performance in all settings with the largest margins observed in the fully unaligned scenario. In contrast, most baselines show weaker performance due to inconsistent view alignment, unreliable sample matching or insufficient information utilization.

5.3.2 Advantages of Cluster-level Manifold Matching via Sample-to-Cluster Discrete Transport

In MVC tasks, consistency alignment is designed to build discriminative semantic boundaries for clustering decision, rather than pursuing high similarity between instance-level representations, view-level distributions or structures. Thus, cluster-level consistent alignment is more compatible with clustering objectives.

Quantitative Analysis. In Table 3, performance differences between manifold alignment methods validate the rationality of sample-to-cluster discrete transport. Most sample-to-sample methods exhibit inferior clustering performance, as excessive smoothing and poor resistance to unaligned noise jointly destroy intrinsic semantic boundaries and cluster structures. Via sample-to-distribution semantic alignment, SMILE alleviates the limitation of pairwise over-smoothing to outperform most baselines, which highlights the compatibility between cross-view alignment levels and many-to-one clustering objectives. Without paired prior for

TABLE 2: Taxonomy and Summary of Comparative Algorithms.

Algorithm	View Consistency Alignment		Sample Correspondence Matching		Assumption Dependency	Model Function Adaptation		Applicable Scenarios
	Alignment Mechanism	Alignment Criterion	Matching Mechanism	Matching Granularity		View Scalability	Sample Utilization	
PVC (20 NIPS)	Sample-to-Sample: continuous smoothing via differentiable Hungarian	Instance-level representation similarity	Hungarian matching	Instance-level one-to-one	Dual-assumption: continuous cluster manifold, reliable paired priors	Local consistency: pairwise cross-view alignment, dual-view restricted	Biased modeling: paired samples utilized, unpaired fully discarded	Full aligned Partial aligned
GWMAC (21 KDD)	Sample-to-Structure: continuous smoothing via continuous OT	View-level structural isomorphism	Structure matching via Gromov-Wasserstein OT	Neighborhood-level many-to-one	Single-assumption: continuous cluster manifold	Global consistency: full-view parallel interaction, multi-view scalable	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned
MvCLN (21 CVPR) SURE (2022 TPAMI)	Sample-to-Sample: continuous smoothing via contrastive learning	Instance-level semantic consistency	Nearest neighbor greedy matching	Instance-level one-to-one	Dual-assumption: continuous cluster manifold, reliable paired priors	Local consistency: pairwise cross-view alignment, dual-view restricted	Biased modeling: paired samples utilized, unpaired fully discarded	Full aligned Partial aligned
SMILE (24 TPAMI)	Sample-to-Distribution: continuous smoothing via information-theoretic constraint	View-level distribution consistency	Top-K neighbor matching	Neighborhood-level many-to-one	Single-assumption: continuous cluster manifold	Global consistency: full-view parallel interaction, multi-view scalable	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned
CGCN (24 TCSVT)	Sample-to-Sample: continuous smoothing via contrastive learning	Instance-level neighborhood preservation	Nearest neighbor greedy matching	Instance-level one-to-one	Dual-assumption: continuous cluster manifold, reliable paired priors	Global consistency: full-view parallel interaction triple-view scalable	Biased modeling: paired samples utilized, unpaired fully discarded	Full aligned Partial aligned
RMCNC (24 TKDE)	Sample-to-Sample: continuous smoothing via contrastive learning	Instance-level representation similarity	Nearest neighbor greedy matching	Instance-level one-to-one	Dual-assumption: continuous cluster manifold, reliable paired priors	Local consistency: pairwise cross-view alignment, dual-view restricted	Unbiased modeling: all samples utilized, paired priors required	Full aligned Partial aligned
VITAL (24 MM)	Sample-to-Distribution: continuous smoothing via generative-contrastive learning	View-level distribution consistency	Nearest neighbor greedy matching	Neighborhood-level one-to-one	Single-assumption: continuous cluster manifold	Local consistency: pairwise cross-view alignment, dual-view restricted	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned
CANDY (24 NIPS)	Sample-to-Sample: continuous smoothing via contrastive learning	Instance-level semantic consistency	Nearest neighbor matching	Instance-level one-to-one	Single-assumption: continuous cluster manifold	Local consistency: pairwise cross-view alignment, dual-view restricted	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned
TCLPVC (25 TCSVT)	Sample-to-Sample: continuous smoothing via contrastive learning	Instance-level representation similarity	Self-paced semantic matching based on Hungarian strategy	Neighborhood-level one-to-one	Single-assumption: continuous cluster manifold	Local consistency: pairwise cross-view alignment, dual-view restricted	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned
MvDOT	Sample-to-Cluster: discrete matching via discrete OT	Cluster-level semantic-structural consistency matching	Cluster information matching via semantic-topological OT	Cluster-level many-to-one	Assumption-free: discrete non-convex structures, unreliable paired priors adaptive	Global consistency: full-view parallel interaction, multi-view scalable	Unbiased modeling: all samples utilized without paired priors	Full aligned Partial aligned Full unaligned

TABLE 3: Performance comparison on six benchmarks. The best and second-best results are highlighted in red and blue.

Aligned Rate	Competitive Method	CUB			Caltech-5V			NUS-WIDE			ALOI-100			Reuters			NoisyMNIST		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
0.0	PVC [41]	46.83	53.24	33.93	62.28	59.77	48.11	10.00	0.00	0.00	40.16	64.89	27.60	38.03	13.96	13.31	Out of Memory		
	GWMAC [48]	43.01	45.86	28.61	70.28	61.37	52.83	17.71	12.71	4.82	10.19	53.64	9.90	27.90	7.48	4.13	23.47	13.72	7.65
	MvCLN [42]	46.50	44.75	32.18	60.56	43.78	37.72	30.97	12.98	8.37	28.31	47.85	16.96	34.55	16.50	12.70	22.37	13.12	9.13
	SURE [43]	48.50	45.47	31.24	69.57	59.31	53.05	22.26	9.51	5.01	36.04	59.95	24.01	35.03	14.57	13.52	28.59	19.90	19.90
	SMILE [26]	62.70	61.76	47.62	74.14	61.36	54.45	48.78	38.40	31.31	28.55	53.20	18.10	38.43	14.85	13.95	10.66	0.03	0.00
	CGCN [39]	58.33	56.23	43.47	82.50	72.96	69.20	45.43	31.95	24.74	38.35	60.14	26.27	36.20	18.06	12.93	33.96	27.81	16.93
	RMCNC [21]	61.33	58.02	46.00	74.21	65.82	55.46	32.52	17.93	11.64	33.91	58.95	23.89	39.65	15.18	11.18	29.09	22.56	13.33
	VITAL [29]	44.83	42.88	27.45	42.16	38.99	25.67	33.73	23.65	14.92	39.16	63.95	27.64	38.87	15.24	7.88	48.34	49.07	33.01
	CANDY [61]	41.47	36.19	28.72	66.14	52.63	49.61	47.22	34.73	25.97	42.20	68.59	35.63	36.29	17.75	15.42	35.20	25.91	13.41
	TCLPVC [5]	43.33	46.03	28.85	62.93	51.80	43.72	30.73	17.94	11.30	35.14	58.10	22.66	43.79	28.68	17.82	Out of Memory		
	MvDOT-2V	/	/	/	84.64	73.91	70.65	/	/	/	53.68	72.54	38.59	39.21	15.15	14.56	/		
	MvDOT	67.00	62.51	49.14	86.57	76.49	74.14	53.51	38.57	31.59	58.56	75.00	45.21	50.68	33.16	23.46	63.04	53.74	46.16
0.5	PVC [41]	53.87	58.93	41.96	70.93	61.16	52.58	10.00	0.00	0.00	41.21	65.59	30.06	42.06	17.69	12.50	Out of Memory		
	GWMAC [48]	44.50	46.01	28.42	70.50	60.91	52.46	17.68	12.14	4.67	34.76	66.36	17.72	29.09	6.54	4.72	27.68	18.75	10.89
	MvCLN [42]	53.50	51.98	38.51	63.71	48.07	45.03	31.63	13.84	8.46	42.13	65.56	31.36	42.84	21.96	17.92	87.46	73.91	74.55
	SURE [43]	54.51	51.32	37.99	75.36	66.19	60.58	53.68	40.22	32.17	59.28	71.20	44.18	41.72	28.57	22.31	88.22	77.51	77.37
	SMILE [26]	64.83	70.68	59.41	87.21	77.89	75.92	56.74	47.32	44.65	26.81	53.54	16.79	41.59	14.93	14.31	54.90	24.89	27.06
	CGCN [39]	46.50	49.24	36.79	80.79	69.12	65.67	49.22	40.22	30.83	37.41	63.93	23.26	42.90	20.32	17.69	61.67	53.38	45.89
	RMCNC [21]	66.83	63.86	49.82	73.29	62.09	56.17	61.18	47.53	42.53	39.46	63.21	28.03	48.61	31.36	21.15	95.28	87.91	90.06
	VITAL [29]	78.30	75.25	65.05	75.63	64.92	59.44	63.02	48.15	43.10	65.06	78.00	53.66	44.09	22.75	17.31	96.17	91.53	92.00
	CANDY [61]	51.08	47.45	35.16	74.79	59.33	55.76	58.30	43.32	35.53	65.54	81.31	56.92	44.96	24.68	22.19	50.05	41.41	29.11
	TCLPVC [5]	51.67	53.02	37.03	73.43	69.00	61.45	43.50	33.41	25.03	43.83	63.34	30.53	50.76	32.11	25.72	Out of Memory		
	MvDOT-2V	/	/	/	86.79	78.90	75.64	/	/	/	66.22	79.43	54.44	51.45	33.46	28.16	/		
	MvDOT	77.67	73.08	62.35	89.21	79.80	78.60	65.78	48.90	44.84	81.44	87.25	72.24	54.65	33.57	29.76	97.60	93.36	94.81
1.0	PVC [41]	59.31	66.09	52.28	74.29	67.39	58.58	10.00	0.00	0.00	45.44	67.57	31.94	47.98	26.15	21.07	Out of Memory		
	GWMAC [48]	46.50	42.63	25.88	71.14	62.15	53.73	25.82	19.95	9.11	37.65	67.85	20.29	31.13	8.90	5.82	30.17	20.72	13.34
	MvCLN [42]	59.20	55.64	42.15	79.36	69.47	63.64	51.60	41.14	32.88	47.42	72.83	33.76	46.72	29.23	23.38	98.47	95.55	96.65
	SURE [43]	61.83	54.96	41.61	81.93	69.34	65.58	56.71	40.44	34.24	63.02	75.18	49.22	52.37	30.39	26.86	95.81	89.35	90.99
	SMILE [26]	72.17	79.57	69.66	88.18	78.93	77.37	58.77	51.39	46.58	35.99	58.57	24.49	43.54	17.60	15.59	99.36	98.45	97.86
	CGCN [39]	78.33	75.59	64.32	82.43	72.37	69.10	56.39	45.12	35.48	60.78	78.10	44.81	47.96	28.83	23.11	85.34	72.02	71.44
	RMCNC [21]	72.67	66.74	54.53	80.57	73.17	62.23	67.32	53.00	48.23	42.15	65.82	30.37	51.11	33.35	26.77	98.88	96.61	97.56
	VITAL [29]	85.03	78.60	72.61	80.86	67.42	63.05	66.71	52.85	48.76	73.73	85.19	70.61	49.08	25.18	20.69	98.22	96.25	97.07
	CANDY [61]	72.00	65.67	55.36	80.64	68.08	65.01	62.43	49.74	39.39	69.90	84.21	61.98	54.46	34.43	29.32	65.75	54.32	47.84
	TCLPVC [5]	55.50	57.45	40.65	79.29	74.79	67.02	56.69	42.57	35.29	57.36	75.15	46.15	54.21	36.81	31.76	Out of Memory		
	MvDOT-2V	/	/	/	89.14	79.69	78.60	/	/	/	71.23	85.82	65.23	54.42	31.55	26.68	/		
	MvDOT	87.33	79.87	74.89	89.36	83.27	80.29	66.97	53.42	49.19	86.29	92.12	80.22	58.87	37.78	35.58	99.54	98.51	98.99

alignment guidance, consistency smoothing over view distributions may collapse data structures from discrete non-convex manifolds with diverse and separable semantics into continuous convex

manifolds with sparse and ambiguous semantics, while lacking structural modeling to maintain or recover cluster boundaries. These issues restrict SMILE on datasets with narrow inter-cluster

margins and massive pseudo-semantic noise, like ALOI-100 and NoisyMNIST. GWMAC focuses on structural isomorphism but ignores explicit semantic constraints, which fails to rectify semantic mismatch induced by misalignment, as evidenced by degraded performance on NUS-WIDE with semantic gaps and ALOI-100 with semantic diversity. By unifying cross-view matching and clustering within a sample-to-cluster map with joint semantic-structural constraints, MvDOT alleviates cluster structure degradation caused by similarity smoothing over false-positive pairs.

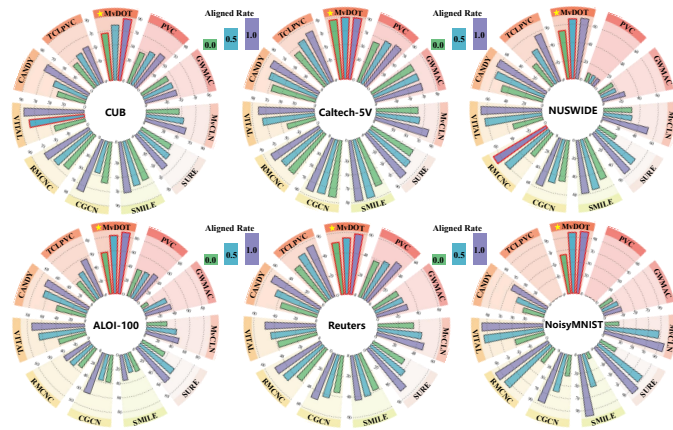


Fig. 4: Visualization for Table 3 on metric ACC.

Qualitative Visualization. To further show MvDOT's ability in preserving clear semantic boundaries and discrete non-convex structures, in Fig. 5, we adopt the scatter heat map to visualize clustering results, together with their inter-cluster density curves at an aligned rate of 0.5. Sample-to-sample methods yield abundant inter-cluster gaps and sparse intra-cluster density, accompanied by frequent misclassifications. Such blurred boundaries and loose structures arise from excessive continuous smoothing, which improperly interpolates pseudo-semantic samples to connect semantically and structurally inconsistent manifolds. Specifically, pointwise similarity constraints loosen cluster boundaries and weaken cluster cohesion, while prototype-based distribution alignment in SMILE preserves intra-cluster compactness but compresses inter-cluster gaps by pulling prototypes closer for consistency alignment. Consequently, SMILE performs well on view-heterogeneous NUS-WIDE but tends to suffer semantic confusion on cluster-diverse ALOI-100. By contrast, MvDOT employs the OT barycenter as a shared alignment anchor to transport cross-view samples in parallel toward barycenter-anchored consensus clusters. Our sample-to-cluster discrete transport avoids pseudo-semantic interpolation and restore data structures with high intra-cluster density and low inter-cluster density. **Robustness.** As the aligned rate decreases from 1.0 to 0.0, MvDOT exhibits smooth and controlled performance decline within approximately 30%, while most baselines suffer sharp drops. This indicates that MvDOT can effectively prevent cluster collapse from cross-view correspondence degradation.

5.3.3 Advantages of Cluster-level Sample Matching via Many-to-One Optimal Transport

In UMVC tasks, sample matching aims to provide reliable consistency information to improve clustering performance, rather than enforcing strict one-to-one correspondences as in retrieval tasks. Accordingly, cluster-level soft matching is more flexible and robust for clustering tasks.

Quantitative Analysis. Performance differences across sample matching methods in Table 3 validate the superiority of relaxed correspondences at semantic-level and cluster-level, e.g., SMILE, TCLPVC and MvDOT. Although supporting soft matching, SMILE merely conducts neighborhood-level local matching without global semantic constraints, which introduces distribution bias to bring about noticeable performance drop on large-scale noisy datasets, like NoisyMNIST, as unaligned rate increases. By incorporating semantic self-paced filtering into Hungarian matching, TCLPVC excels on cluster-scale imbalanced Reuters, as semantically consistent correspondences are prioritized to stabilize self-supervision and mitigate large-cluster dominance. However, its one-to-one constraint renders it vulnerable to unreliable similarity estimation on view heterogeneous NUS-WIDE or semantic diverse ALOI-100, where early mismatches tend to be amplified. MvDOT adopts the OT barycenter to adaptively perceive and transport cluster information across views for matching guidance. As a shared matching reference, the barycenter provides unbiased semantic information to guarantee global semantic consistency, while imposing isomorphic topological constraints to further regularize local structural consistency.

Qualitative Visualization. To intuitively demonstrate MvDOT's superiority in recovering cluster-level correspondences, we adopt Sankey diagrams to visualize matching results under 50% misalignment on Caltech-5V and Reuters, as shown in Fig. 6. Ideally, correct alignment should manifest as clear and concentrated one-to-one flow between the same cluster across views. As illustrated, MvDOT produces the cleanest and most interpretable cross-view flows, with dominant connections concentrated between semantic corresponding clusters. SMILE and TCLPVC recover partially correspondences but still suffer from noticeable cross-cluster leakage. In contrast, one-to-one methods display heavily fragmented flows due to mismatched correspondence recovery. Notably, CANDY exhibits a clear class-wise mass imbalance between two views, especially on Reuters. This is mainly caused by its nearest-neighbor matching without exclusivity or de-duplication constraints, where the same sample in View 1 can be repeatedly selected by multiple samples in View 0. Such replacement-style matching may hurt the semantic diversity of the reordered View 1 and introduce cross-class mismatch noise. In over-smoothed representation spaces, samples from dense large clusters are more likely to be repeatedly selected as nearest neighbors, while small clusters are under-matched, which leads to a typical large-cluster absorbing small-cluster phenomenon and an imbalanced class distribution after reordering.

These observations confirm the rationality of our cluster-level matching. Reliable correspondence recovery requires semantic similarity and topological connectivity constraints to suppress pseudo-semantic noise and preserve inherent cluster structures.

5.3.4 Advantages of Full-view Information Utilization Without Assumption Dependence

Most methods in Table 2 rely on strong assumptions like continuous cluster manifolds and reliable paired samples. Accordingly, local interactions among limited pairwise across heterogeneous views cause irreversible information loss and biased clustering assignment, as evidenced by poor performances in Table 3.

Driven by pairwise interactions, sample-to-sample methods struggle to fairly model global consistency across all views in a parallel manner. Though CGCN supports multi-view interaction, it discards unpaired but similar neighbors to construct positive pairs,

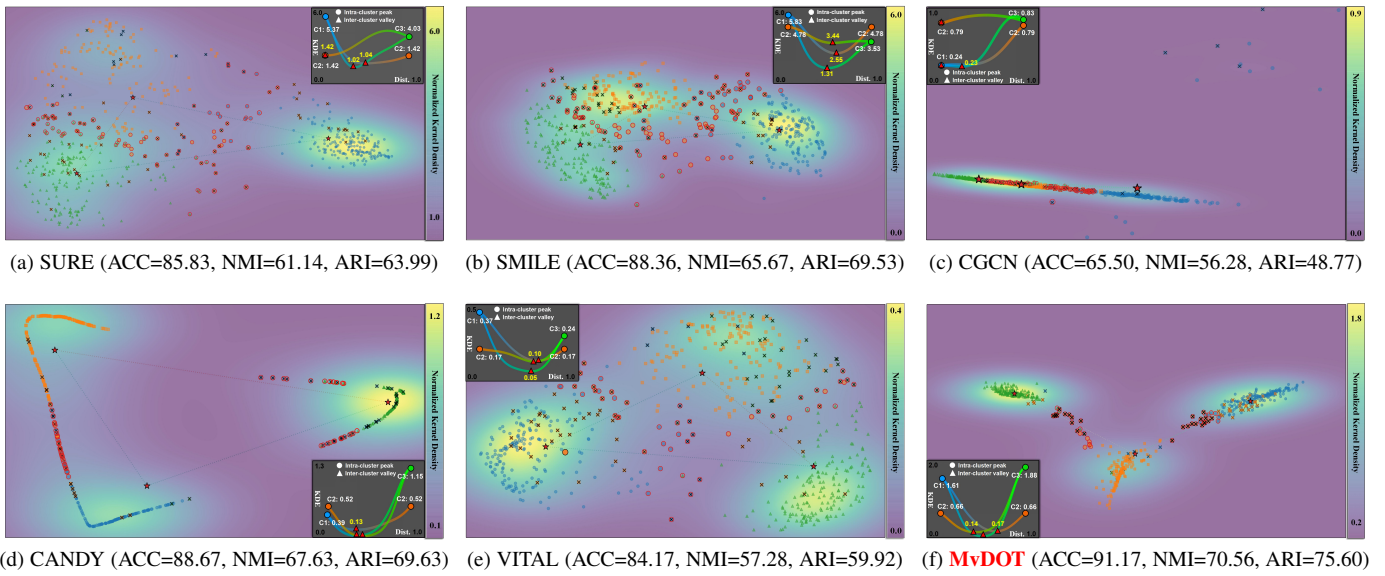


Fig. 5: Clustering visualization on the Synthetic3d dataset (300 samples, 3 views, 3 classes) at an aligned rate of 0.5. A shared PCA projector embeds samples into 2D space to illustrate the global distribution. (a) Scatter heatmap: Samples are annotated with distinct colors and markers based on ground-truth labels (K-means cluster centroids “★”, misclassified samples “×”, inter-cluster gap samples “○”). A kernel density map normalized by Gaussian noise is overlaid to intuitively visualize clustering quality. Fewer misclassified/gap samples, higher intra-cluster density, and larger inter-cluster margins indicate weaker pseudo-semantic interference and over-smoothing, yielding clearer cluster structures. (b) Kernel density curves: Inter-cluster density variation indirectly characterizes clustering quality. Each curve contains two intra-cluster density peaks and one inter-cluster density valley; higher peaks and lower valleys (forming a pronounced concave curve) indicate superior intra-cluster compactness and inter-cluster separability, while flatter curves (even with convex peaks) reflect looser structures and weaker separation.

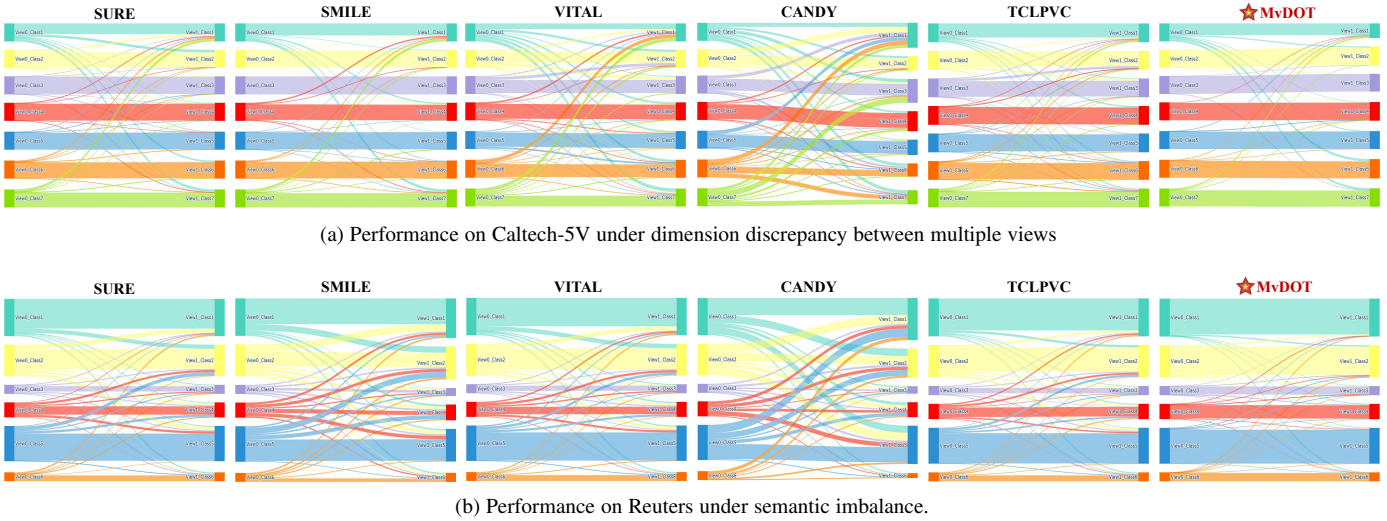


Fig. 6: The Sankey diagram shows cross-view correspondences recovered by different methods under an aligned rate of 0.5. Two columns represent two views, and different colors denote different clusters. Ideally, cross-view matching is reflected by a clear one-to-one connection between the same cluster across views. The dispersion and crossing of the links reflect the accuracy and stability of the matching: more concentrated links with fewer crossings imply more accurate correspondences; conversely, the matching suffers from worse semantic confusion or alignment bias.

which introduces both biased information and false negatives. This explains why CGCN performs better under fully unaligned settings than partially aligned settings in Table 3, since the former avoids aggressive pairwise filtering and uses full sample information to discover cluster structures, albeit with more false positives. To alleviate these issues, VITAL models the mean-variance statistics of multi-view distributions to mine consistency and complementary information, while designing dynamic rectification to reduce false negatives. Its effectiveness implicitly assumes that

same-cluster manifolds across views share semantic overlapping to form continuous distributions, so that their mean-variance can capture a convergent central tendency. On view-heterogeneous datasets like NUS-WIDE, its same-cluster manifolds are dispersed and their mean centers degenerate into semantic compromises instead of unbiased anchors to guide consistency alignment. Under unaligned noise, dynamic rectification fails to rectify unreliable similarity measures and accumulate erroneous matches.

MvDOT relaxes assumptions of continuous distribution and

TABLE 4: Ablation study on different loss terms, alignment anchor, and matching strategy.

Ablation Component	Aligned Rate	CUB			Caltech-5V			NUS-WIDE			ALOI-100			Reuters			NoisyMNIST		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
MvDOT	0.0	67.00	62.51	49.14	86.57	76.49	74.14	53.51	38.57	31.59	58.56	75.00	45.21	50.68	33.16	23.46	63.04	53.74	46.16
	0.5	77.67	73.08	62.35	89.21	79.80	78.60	65.78	48.90	44.84	81.44	87.25	72.24	54.65	33.57	29.76	97.60	93.36	94.81
	1.0	87.33	79.87	74.89	89.36	83.27	80.29	66.97	53.42	49.19	86.29	92.12	80.22	58.87	37.78	35.58	99.54	98.51	98.99
W/O $\mathcal{L}_{sm}$	0.0	65.00	63.80	48.89	83.07	72.55	69.08	48.19	35.87	28.68	51.79	72.32	39.04	45.30	23.26	19.41	53.99	45.93	35.75
	0.5	64.50	59.61	45.85	85.14	75.39	71.59	51.06	37.91	31.12	61.67	81.23	50.15	47.81	29.06	23.91	69.67	88.91	72.36
	1.0	69.17	72.09	59.30	86.57	78.22	74.90	54.64	42.49	24.46	77.45	93.03	76.27	51.24	31.19	25.69	84.97	94.14	86.61
W/O $\mathcal{L}_{mm}$	0.0	54.83	54.07	36.05	81.71	72.43	67.76	45.76	30.89	24.43	46.66	68.71	36.18	40.63	18.11	15.89	49.22	40.70	31.21
	0.5	70.83	70.74	56.50	83.96	71.55	68.08	57.72	42.93	36.40	78.11	87.49	69.25	50.47	31.41	26.24	96.12	90.05	91.66
	1.0	73.67	71.80	59.99	87.71	80.90	76.87	61.28	47.31	42.25	81.59	94.83	81.08	54.55	35.31	29.91	98.66	97.64	98.25
Replace OT Barycenter with K-Means Centroids	0.0	55.50	53.99	38.63	68.71	69.25	58.95	48.51	33.60	26.54	52.69	74.79	42.96	49.03	25.83	22.15	58.39	47.96	39.69
	0.5	60.67	70.33	52.75	72.50	72.68	61.99	57.93	45.31	38.49	71.36	85.58	64.28	51.28	30.66	25.95	87.47	89.27	84.57
	1.0	77.00	76.16	65.25	77.36	81.59	72.53	59.81	50.49	44.39	75.50	89.50	67.91	54.78	34.38	32.03	91.81	95.93	90.76
Replace Cluster-level Matching with Greedy Matching	0.0	63.67	54.67	41.20	81.64	70.60	66.57	46.10	34.54	26.38	47.25	67.22	35.25	38.53	14.19	11.63	50.91	42.89	33.13
	0.5	73.17	66.02	55.03	82.86	72.12	68.60	58.21	43.39	37.76	67.11	74.33	51.52	51.18	28.93	25.26	87.56	85.85	83.09

paired prior. By integrating full-view information into a global semantic-geometric center, its OT barycenter serves as an unbiased reference for global consistency alignment, which promotes information interaction from dual-view paired samples covering multi-view full samples. MvDOT achieves stable and superior NMI on Reuters as the alignment rate decreases, because the OT barycenter captures cluster semantics and scales from the empirical distribution and provides unbiased information for clustering guidance, to prevent sparse clusters from being overwhelmed by dominant large clusters. Moreover, MvDOT-2V remains competitive even with two views in Table 3 (on two-view datasets, MvDOT-2V performs identically to MvDOT, denoted by "V"), while full-view utilization further boosts performance. It highlights MvDOT's robustness under limited information and the necessity of unbiased information utilization and global consistency alignment.

5.4 Understanding for MvDOT

5.4.1 Ablation Study

To explore the core factors behind MvDOT's strong adaptability to complex scenarios, we design four ablation experiments to analyze the roles of the discrete manifold transport loss $\mathcal{L}_{mm}$ and unpaired sample matching loss $\mathcal{L}_{sm}$, and evaluate performance variations when replacing the OT barycenter with the K-means centroid and semantic-topological transport with greedy matching.

Table 4 shows that both barycenter-guided manifold learning and sample matching contribute substantial performance. In fully and partially aligned settings, $\mathcal{L}_{sm}$ benefits from available paired priors or high-confidence sample correspondences recovered by our semantic-topological OT, which can provide stronger self-supervised signals to achieve larger improvements than $\mathcal{L}_{mm}$. Once paired priors are completely absent, $\mathcal{L}_{mm}$ exhibits superior robustness than $\mathcal{L}_{sm}$, as its OT barycenter can transport all samples to their consensus clusters under consistent semantic-geometric constraints. Moreover, MvDOT equipped with $\mathcal{L}_{mm}$ yields more stable cluster structures, as validated by higher NMI scores. This confirms that sample-to-cluster transport aligns and merges cross-view discrete manifolds to strengthen compact structures while maintaining separable boundaries.

Furthermore, using the K-means centroid leads to significant performance degradation, as first-order statistics collapse intra-cluster semantic-geometric variations into positional information and cannot model higher-order cluster patterns. In contrast, our

OT barycenter jointly optimizes semantic-structural transport costs across all views to capture consensus cluster information as an unbiased guidance for global consistency clustering. Meanwhile, our semantic-topology transport performs cluster-level soft matching by globally searching for cross-view correspondences with consistent semantic assignment and isomorphic topological connectivity, thereby suppressing erroneous connections from unaligned noise.

5.4.2 Convergence Analysis

As shown in Fig. 7, we conduct convergence experiments on two representative datasets: semantically diverse ALOI-100 and large-scale NoisyMNIST. All clustering metrics rapidly increase and smoothly converge to steady values with minor fluctuations. To analyze the coupled optimization dynamics, we further decompose $\mathcal{L}_{mm}$ into the multi-view transport term $\mathcal{L}_{mot}$ and the graph regularization term $\mathcal{L}_{gr}$, i.e., $\mathcal{L}_{mm} = \mathcal{L}_{mot} + \mathcal{L}_{gr}$ according to Eq. 30. The core losses $\mathcal{L}_{mot}$ and $\mathcal{L}_{sm}$ decrease monotonically, confirming that barycenter-guided discrete manifold transport and unpaired sample matching are jointly optimized in a coupled manner. The low and stable values of $\mathcal{L}_{rec}$ and $\mathcal{L}_{gr}$ indicate that view-specific semantic information and neighborhood structures are consistently continuously preserved to prevent semantic space and cluster structure collapse caused by cross-view over-smoothing.

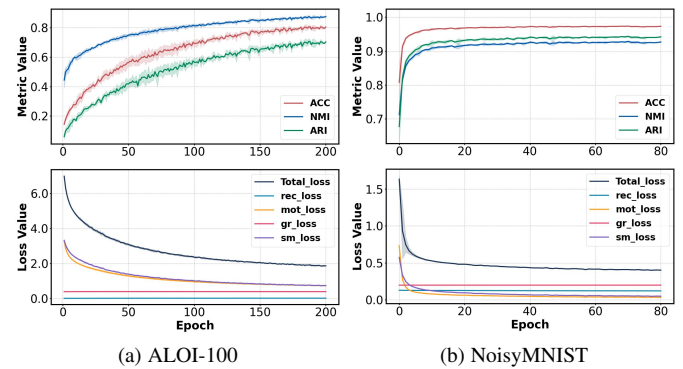


Fig. 7: Learning curves of MvDOT under an aligned rate of 0.5. All curves are averaged over 5 random trials with error bands.

Although coupled multi-tasks are optimized within an adaptive end-to-end framework, the performance metrics improve consistently as the losses converge, and the training process does not

suffer from obvious oscillation or divergence. This provides solid evidence for the design rationality of our equal-weight loss and the optimization stability of MvDOT.

5.4.3 Sensitivity Analysis

To prevent model performance collapse from hyperparameter variations, we focus on two key hyperparameters for sensitivity analysis: the update momentum α of the barycenter weight in Eq. 33, and the graph regularization coefficient λ for the structure constraint in Eq. 30. To ensure experimental validity, we adopt broad but reasonable value ranges: $\alpha \in [0.80, 0.98]$ and $\lambda \in [0.05, 1.0]$.

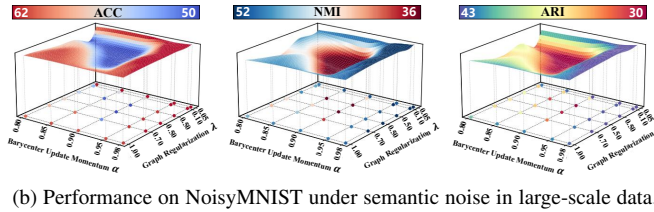
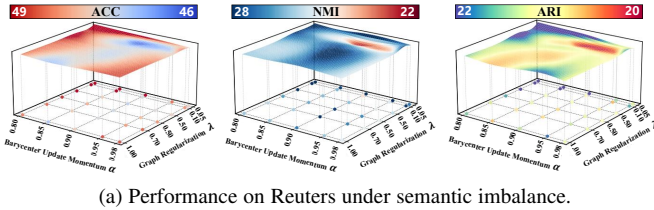


Fig. 8: Parameter analyses for the barycenter update momentum α and the graph regularization coefficient λ under an aligned rate of 0.0.

Fig. 8 shows that MvDOT does not suffer severe performance degradation under reasonable hyperparameter variations, even on Reuters and NoisyMNIST with complex semantic distributions. Compared with α , MvDOT is more sensitive than λ , as λ strictly regulates the strength of structural preservation during consistency alignment. An overly small λ imposes weak structural constraints, whereas an overly large λ permits local topological relations to dominate the learning, which restricts the exploration and utilization of global semantic information. In fully unaligned scenarios with abundant pseudo-semantic noise, overreliance on structural alignment may aggravate semantic confusion without learning reliable semantic guidance. Accordingly, we set $\lambda \in \{0.05, 0.1\}$, which keeps semantic consistency as dominant guidance for optimization and employs structural isomorphism as an auxiliary regularizer to ensure alignment effectiveness. To facilitate stable barycenter evolution, we prefer a large momentum $\alpha \in \{0.95, 0.98\}$ to mitigate optimization bias inherent to unaligned noise and further amplified by batch-wise sampling.

5.4.4 Visualization Clustering

To intuitively present the cluster partition capability of MvDOT, we plot similarity matrices and t-SNE visualizations based on consensus representations $\mathbf{H}$ and ground-truth labels $\mathbf{Y}$. Fig. 9 verifies cluster semantic discriminability from the representation similarity perspective. Even on semantic diverse and imbalanced datasets, their similarity matrices exhibit clear block-wise structures, with strong intra-cluster similarity and inter-cluster discrepancy. This indicates that our sample-to-cluster transport matching enhances inter-cluster semantic discriminability. By adopting clustering objectives to guide consistency learning, it avoids over-smoothing of instance or view representations, which increases

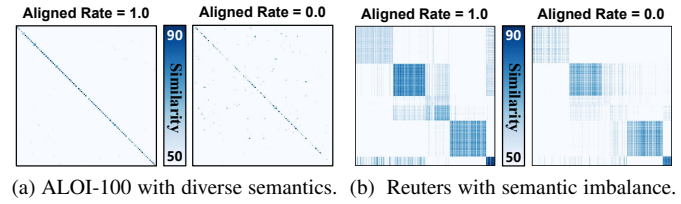


Fig. 9: Similarity matrices of consensus representations $\mathbf{H}$.

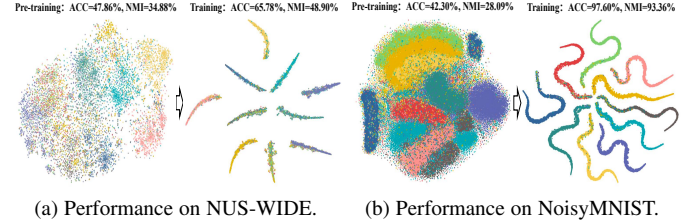


Fig. 10: Clustering structure visualization under aligned rate 0.5.

clustering uncertainty due to semantic boundary degradation. Even under fully unaligned settings, MvDOT is able to restore clear cluster structures by integrating semantic-structural information across all views. Fig. 10 further evaluates cluster structure quality from a geometric stability perspective. On view heterogeneous or large-scale noisy datasets, MvDOT achieves minimal misclassification while maintaining intra-cluster compactness and inter-cluster separation. As observed, MvDOT can learn clustering-friendly representations to recover robust cluster structures. Even with unaligned noise, distinct semantic clusters avoid erroneous connection or merging caused by pseudo-semantic interpolation.

Both Fig. 9 and Fig. 10 show MvDOT recovers compact geometric structures with clear semantic boundaries under unaligned noise. These benefits from adopting clustering objectives to guide consistency learning, MvDOT avoids over-smoothing of instance or view representations, which increases clustering uncertainty due to semantic boundary degradation. Furthermore, by jointly modeling semantic similarity and topological connectivity as cluster-level consistency constraints, MvDOT resists unaligned noise to recover reliable cluster relationships.

6 CONCLUSION

Constrained to smoothing cross-view representations, distributions, and structures with high similarity for consistency clustering, existing MVC methods tend to blur semantic boundaries and degrade cluster structures under semantic and instance view-unaligned noise. To tackle these limitations, we novelly extend Semi-Discrete OT to the MVC setting, and jointly formulate cross-view alignment and clustering as a sample-to-cluster transport matching, namely MvDOT. Free from view-aligned priors, MvDOT globally captures view-shared semantic-geometric information to learn an OT barycenter with well-separated cluster regions in the common space, and unbiasedly transports view-specific samples to their corresponding consensus clusters. By imposing both global semantic consistency and local topological constraints on sample-to-cluster transport matching, MvDOT recovers cluster-level cross-view correspondences for consistency clustering. Both theoretical analysis and extensive experiments consistently verify that MvDOT effectively and robustly handles complex MVC tasks.

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